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Master Thesis

生成式人工智慧於數位閱讀沉浸式推薦之應用

Generative AI for Immersive Recommendation in Digital
Reading

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摘 要

隨著數位閱讀與行動裝置的普及，現代讀者的閱讀行為逐漸由專注式、連續性閱讀，轉變為碎片化、情境多變且認知負荷更高的模式。在注意力經濟的脈絡下，傳統以歷史偏好為基礎的書籍推薦系統，面臨無法即時捕捉使用者情境與心理需求的挑戰，難以有效促進閱讀沉浸體驗。

本研究提出一套生成式人工智慧驅動之沉浸式書籍推薦架構，針對行為變遷與碎片化閱讀特性，整合語意理解、情感推理與社群互動等多元訊號，重新設計推薦邏輯。透過閱讀參與指數（REI）、數位格式指數（DFI）與評論情感分數（ReviewScore）等特徵，結合學習排序策略與XGBoost、SVM、大型語言模型（LLM）等方法進行實證驗證。

結果顯示，本架構於NDCG@3、MRR@3、Top-1準確率等指標皆優於傳統方法，驗證生成式AI於行為變動與情境適應層面的優勢，為數位閱讀時代的推薦系統設計提供實質貢獻。

關鍵字：生成式人工智慧, 沉浸式推薦, 碎片化閱讀, 行為變遷, 語意分析, 情感推理, 使用者行為建模

Generative AI for Immersive Recommendation in Digital Reading

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Abstract

With the proliferation of digital reading and mobile devices, modern readers have shifted from focused, continuous reading patterns to fragmented, context-dependent behaviors with increased cognitive load. Under the attention economy, traditional book recommendation systems—relying primarily on historical preferences—struggle to capture users’ real-time contexts and psychological needs, limiting their ability to foster immersive reading experiences.

This study proposes a Generative AI-driven immersive book recommendation framework that addresses behavioral shifts and fragmented reading habits by integrating semantic understanding, emotional inference, and social interaction signals. Novel feature indicators, including the Reading Engagement Index (REI), Digital Format Index (DFI), and Review Sentiment Score (ReviewScore), are incorporated alongside learning-to-rank strategies and models such as XGBoost, SVM, and Large Language Models (LLMs) for empirical evaluation.

Results show that the proposed framework outperforms traditional methods in NDCG@3, MRR@3, and Top-1 accuracy, demonstrating the effectiveness of Generative AI in modeling behavioral variability and contextual adaptation. This research contributes to the design of recommendation systems in the digital reading era.

Keywords: Generative Artificial Intelligence, Immersive Recommendation, Fragmented Reading, Behavioral Shifts, Semantic Analysis, Emotional Inference, User Behavior Modeling.

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Chapter 1. Introduction

1.1 Background

With the rise of digital reading and the widespread influence of social media, modern readers are undergoing a fundamental transformation in their reading behavior [1, 2]. Unlike the linear and immersive reading patterns of the past, today's information consumption is increasingly fragmented, multitasked, and surface-level [3]. Within the framework of the attention economy, where user attention has become a scarce and valuable resource, book recommendation systems face unprecedented challenges [4].

Readers today encounter two key obstacles: first, they often struggle to find books that genuinely match their evolving interests; second, even when books are selected, sustaining focus and engagement is difficult, resulting in disrupted or abandoned reading experiences [5, 6]. These tendencies are well explained by the Cognitive Miser Theory, which posits that individuals prefer to minimize cognitive effort when making decisions—leading them to rely on heuristics like popularity rankings, star ratings, or peer reviews rather than actively exploring new titles [7].

However, traditional recommendation systems, which rely heavily on historical user preferences and static content similarity, lack the capability to adapt to these behavioral and environmental changes [8, 9]. They often fail to capture dynamic shifts in user attention, contextual relevance, or emotional engagement—factors that are increasingly critical in fostering meaningful reading experiences in the digital age.

In this context, generative artificial intelligence (Generative AI) presents a new opportunity. With capabilities in semantic understanding, emotion recognition, and content generation, generative AI models can deeply model reader behavior, book characteristics, and community interactions—thereby producing recommendations with stronger emotional resonance and semantic alignment [10, 11]. Furthermore, the interactive nature of generative AI offers the

potential to shift the role of recommendation systems from static filters to personalized reading guides, actively facilitating sustained attention and deeper engagement.

Accordingly, this study focuses on rethinking book recommendation mechanisms from the perspective of rebuilding immersive reading. By leveraging generative artificial intelligence, we aim to develop a recommendation framework that directly addresses the challenges posed by fragmented attention and declining reading continuity in digital environments.

1.2 Motivation and Research Problems

The widespread adoption of digital devices and mobile platforms has fundamentally altered how individuals interact with text. Reading time is now highly fragmented, with content consumption characterized by brevity, immediacy, and reduced cognitive engagement [3, 4]. As a result, deep and immersive reading is becoming rarer, posing significant challenges for book recommendation systems that aim to foster sustained engagement.

Traditional recommendation approaches, such as collaborative filtering and content-based filtering, are limited in their ability to address these shifts. While effective in modeling historical preferences, they often fail to adapt to real-time behavioral changes, fluctuating attention spans, and emotionally driven consumption patterns. Moreover, these systems rarely consider the psychological conditions required for immersive reading, focusing instead on static relevance rather than dynamic engagement [12, 13].

The Attention Economy Theory emphasizes that human attention has become the most valuable resource in the digital age [4], while the Cognitive Miser Theory [7] highlights the tendency of individuals to minimize mental effort in decision-making. Together, these theories explain why traditional recommendation systems, which neglect cognitive and emotional dimensions, struggle to maintain user engagement in fragmented reading contexts [14, 15].

In contrast, Generative AI offers capabilities that directly address these limitations. Its ability to comprehend semantics, infer emotions, and model contextual nuances enables the development of recommendation mechanisms that align with modern readers' behaviors and attention patterns [16, 17]. By leveraging Generative AI, recommendation systems can evolve from simple preference-matching tools to dynamic, context-aware guides that facilitate immersive reading experiences [18, 19].

Accordingly, this study addresses the following research questions:

(1) How can generative artificial intelligence enhance readers' immersion in digital reading environments through semantic and emotional understanding?

Generative artificial intelligence (AI) enables the modeling of both semantic structures and emotional undertones within textual data, offering deeper alignment between recommended content and readers' internal states. By analyzing review sentiment and narrative tone, the system can identify books that resonate with readers' mood and psychological needs. Unlike traditional models focused solely on preference matching, generative AI captures latent contextual and affective signals that promote deeper engagement. This enhances the reader's immersive experience by reducing cognitive dissonance and increasing emotional relevance. Ultimately, AI shifts the recommendation role from reactive filtering to proactive cognitive and emotional guidance.

(2) Can the integration of social signals, format preferences, and review sentiment create a recommendation mechanism that better aligns with users' attention patterns?

Yes, the integration of multi-dimensional features such as social validation, digital accessibility, and affective feedback enables the system to better capture users' fragmented attention in real-time. Features like award presence, Digital Format Index (DFI), and sentiment polarity reflect how modern readers navigate content under cognitive load. By aligning recommendations with these lightweight heuristics, the system reduces decision fatigue and improves acceptance likelihood. This mechanism responds to the principles of the Attention Economy and Cognitive Miser Theory by emphasizing immediacy, relevance, and emotional resonance. As a result, recommendations become not just accurate, but also timely and cognitively efficient.

This research positions Generative AI as a solution to the behavioral and contextual limitations of traditional methods, aiming to establish a more adaptive and effective book recommendation framework.

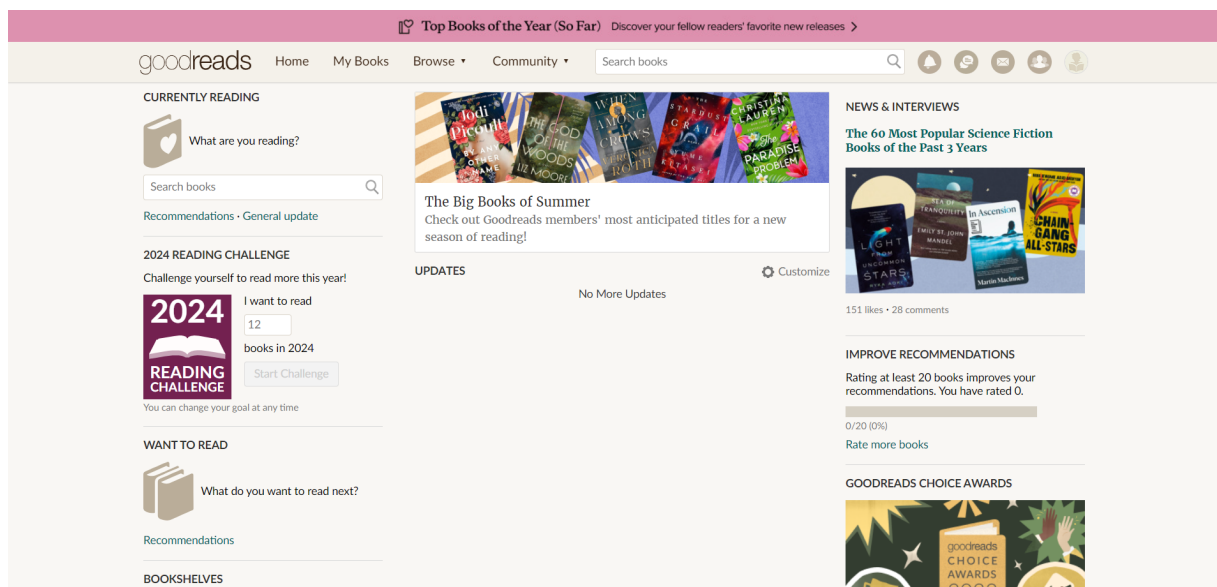


Figure 1: The Snapshot of Goodreads Website

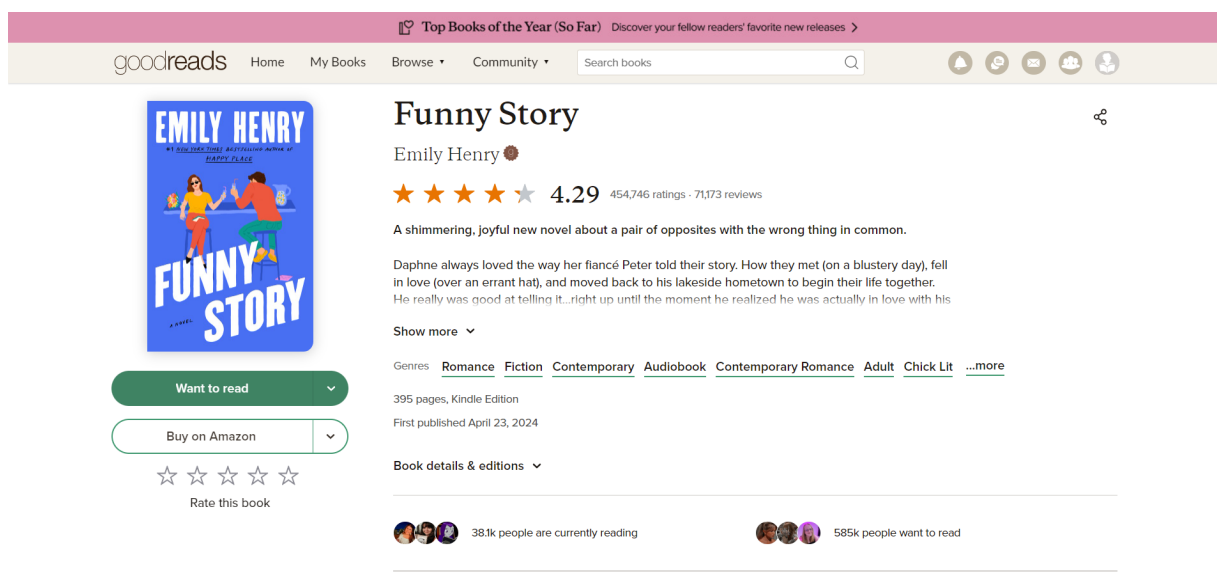


Figure 2: The Snapshot of Goodreads's Book Profile



Figure 3: The Snapshot of Goodreads’s Reader’s Review

1.3 Research Objectives & Contributions

This study aims to develop a personalized book recommendation framework that responds to the behavioral changes and contextual challenges of modern digital reading. Unlike conventional approaches that focus solely on improving prediction accuracy or relevance scores, this research positions recommendation as a form of cognitive and emotional intervention—guiding readers from passive browsing toward deeper and more sustained engagement.

The primary goal is to overcome the inherent limitations of traditional recommendation systems, which often fail to adapt to fragmented attention spans, evolving user behaviors, and affective variability. By leveraging the semantic comprehension, emotional inference, and contextual adaptability of generative artificial intelligence, the proposed framework aspires to deliver recommendations that are not only relevant, but also cognitively lightweight and emotionally aligned with the reader’s current state.

This research focuses on designing a mechanism capable of capturing semantic preferences and emotional tendencies to enhance reading immersion. It further integrates signals such as award visibility, digital format preferences, and sentiment polarity to address the volatility of modern attention patterns. Through empirical evaluation against baseline models such as SVD and BPR, the study demonstrates the superior performance of the generative AI-based approach, especially in fostering immersion under cognitively fragmented conditions.

Contributions of this research include the introduction of novel behavioral and contextual indicators—such as the Reading Engagement Index (REI), Digital Format Index (DFI), and

Review Sentiment Score (ReviewScore)—to better model engagement in digital contexts. The study also develops an Immersion-Oriented Matching Engine, incorporating learning-to-rank strategies across point-wise, pair-wise, and list-wise paradigms, to generate personalized and affectively resonant recommendations. Beyond empirical improvements, the research reframes recommendation systems as tools for supporting cognitive alignment and emotional engagement, offering a theoretical contribution that aligns with the realities of attention scarcity and decision fatigue in the digital era.

Ultimately, this work underscores that Generative AI is not merely a computational advancement, but a critical evolution in recommender system design—one capable of addressing the behavioral complexity and contextual shifts that traditional approaches have long overlooked.

1.4 Thesis Outline

This thesis is organized into six chapters. Chapter 1 introduces the research background, motivation, and objectives, framing the limitations of traditional recommendation systems and the emerging role of Generative AI. Chapter 2 reviews relevant literature in four areas: digital reading behaviors, cognitive theories, media transformation, and the evolution of recommender systems. Chapter 3 outlines the proposed system architecture, describing four core modules that integrate behavioral signals, semantic features, and emotional modeling. Chapter 4 presents the experimental design and feature engineering processes based on Goodreads data. Chapter 5 evaluates the performance of the proposed system using multiple metrics—including ILD—and compares it with traditional methods. Chapter 6 concludes with key findings, limitations, and future research directions.

Chapter 2. Related Literature

2.1 Digital Reading and the Attention Economy

With the widespread adoption of smartphones and social media, the way people engage with information has shifted from concentrated attention to fragmented scanning. According to Wolf and Potter [3], individuals who read primarily through digital platforms tend to exhibit shallower cognitive processing and reduced ability to sustain deep engagement with complex texts. This is echoed by Carr [20], who warns that digital interfaces may rewire the brain to favor quick scanning over deep reading. Furthermore, Qianjing and Lin [11] observe that younger generations, raised in multimedia environments, increasingly prefer interactive formats, real-time responsiveness, and visual stimulation—factors that are reshaping the expectations and behaviors associated with reading.

Within this context, attention has become the most scarce and valuable resource. Davenport and Beck’s [4] theory of the attention economy posits that in an era of information overload, platforms and content creators compete not for information access, but for the limited attention span of users. As Lanham [21] further suggests, the true economic challenge today is not acquiring information, but managing what gets attention. As a result, long-form content such as books finds itself at a disadvantage in this high-speed digital environment, where even milliseconds of delay or irrelevance can result in user abandonment [22].

This transformation imposes two major challenges for recommendation systems. First, recommendations must capture user attention within an extremely short time frame, or risk being ignored. Second, the goal of recommendation is no longer merely to match interests or trigger clicks, but to serve as a cognitive and emotional gateway that draws readers into immersive engagement. Traditional recommender systems, which rely on preference similarity and past behavior, are insufficient in such environments. They lack the ability to model the dynamic and volatile nature of attention, let alone promote long-term reading commitment [23, 24].

To respond to these shifts, recommender systems must begin to incorporate attention modeling as part of their design. Reading recommendation should be viewed not merely as an information filtering task, but as an intervention in readers’ mental and emotional states. Systems that fail to account for users’ short attention spans, impulsive decision patterns, and multitasking behaviors are unlikely to foster meaningful reading engagement [25].

This study positions the design of immersive recommendation systems as a necessary evolution under the attention economy. Rather than optimizing solely for click-through rates or conversion, the system should support a higher-order goal: to reconstruct the mental context for deep and sustained reading. By reframing recommendation as a process of “attracting attention → establishing context → enabling immersion,” we highlight the need for systems that actively support deep engagement. Generative AI plays a critical role in reshaping the function of recommender systems within this new paradigm [10].

In light of these developments, it becomes clear that traditional recommendation systems, which rely primarily on static preference matching, are ill-equipped to capture the transient and emotionally nuanced states of modern readers. This necessitates a more adaptive and context-aware approach, such as the use of Generative AI.

2.2 Media Transformation of Books

In addition to changes in reading behavior, books themselves—as content carriers—have undergone a significant media transformation. From traditional hardcover and paperback formats to e-books, audiobooks, and even interactive digital experiences, today’s readers are presented with a wide array of format choices. This diversification not only reshapes how readers interact with content but also introduces new challenges and opportunities for recommender systems.

One critical insight is that the medium of a book influences its discoverability and accessibility. E-books and audiobooks are increasingly favored by modern readers for their portability, instant availability, and cross-device compatibility. Qianjing and Lin [11] highlight in their study on multimodal learning that when educational content is offered in diverse formats, user attention, acceptance, and engagement increase significantly. This trend is reflected in the publishing industry, where publishers are now proactively releasing electronic and audio versions of their works to meet the needs of mobile and time-constrained readers.

However, most mainstream recommender systems fail to account for format preference. In

collaborative filtering or semantic-based models, a book is often treated as a single content entity, without differentiation between its print, digital, or audio versions. This simplification can result in recommendations that are mismatched with user reading habits or accessibility constraints, ultimately reducing effectiveness and motivation [26, 27].

To address this gap, this study introduces the Digital Format Index (DFI), which measures the proportion of a book’s editions that are available in digital formats such as e-books and audiobooks. By quantifying the book’s availability in these modern media forms, DFI reflects the readability and adaptability of a title in contemporary reading contexts. Incorporating this indicator into recommendation ranking allows the system to offer more format-flexible options, thus closing the gap between recommendation output and real-world reading needs.

Moreover, the shift in medium has also changed how content is experienced. Enhanced features in e-book platforms (e.g., built-in dictionaries, annotation tools, reading speed controls) and the vocal interpretation in audiobooks can strengthen emotional connection and immersion for the reader [3]. If recommender systems can integrate these media-layer sensory enhancements, they are more likely to support deep and focused reading.

Therefore, understanding how books are represented across digital formats is not a peripheral consideration, but a core design component for building recommendation systems that promote immersive reading in the digital age.

2.3 Reading Psychology and Cognitive Behavior

Beyond environmental and media transformations, reading itself is a psychological and cognitive process. In digital reading environments, users are constantly required to evaluate and choose among an overwhelming number of book options. However, human decision-making rarely involves exhaustive rational analysis. According to the Cognitive Miser Theory proposed by Fiske and Taylor [7], individuals tend to conserve cognitive resources by relying on heuristics rather than engaging in effortful processing.

Applied to reading behavior, this theory suggests that modern readers often depend on simple cues such as bestseller rankings, star ratings, well-known authors, or social endorsements when selecting books—rather than thoroughly comparing summaries or reading samples [28]. This tendency is further exacerbated in attention-scarce environments, where users may make “click or skip” decisions in a matter of seconds. In such contexts, recommendation systems that

fail to present cognitively lightweight and trustworthy signals are likely to be ignored.

Additionally, Orbell and Dawes [29] extend this theory by suggesting that decision-making is also shaped by habitual patterns and prior experiences. Readers often return to familiar genres, preferred authors, or socially validated content, avoiding the mental burden of evaluating unfamiliar material. These behaviors highlight the importance of “effort minimization” and social comparison in digital content consumption.

For recommendation systems seeking to promote immersive reading, two directions emerge. First, systems must provide low-friction, cognitively efficient interfaces that simplify decision-making. Second, they should offer progressive diversity, gradually introducing readers to novel content without overwhelming them. Studies have shown that when recommendation systems present explanations in intuitive language—such as through key-point summaries or emotional cues—users are more likely to trust the system and engage with unfamiliar recommendations [30].

In this study, the insights of cognitive miser theory are translated into design principles: recommendation systems should not only present books that match user preferences but also ensure that recommendations are emotionally accessible, cognitively lightweight, and easy to act upon. This is where generative artificial intelligence plays a crucial role: its ability to process semantics and sentiment makes it well-suited to reduce decision complexity and enhance user receptivity, thus increasing the likelihood of immersive reading engagement.

These findings support the view that modern readers require cognitively lightweight, emotionally resonant, and behaviorally aligned recommendation strategies—areas where Generative AI holds promising advantages over traditional models.

2.4 Recommender Evolution and Generative AI

As digital reading behavior evolves, so too must the technologies that support content discovery and engagement. Traditional recommendation systems—such as collaborative filtering and content-based filtering—have long served as the backbone of digital platforms. These methods typically rely on user-item interaction histories or manually extracted features to generate predictions. While effective in structured domains, such systems exhibit clear limitations when applied to dynamic, emotionally nuanced tasks like promoting immersive reading.

One critical drawback is that these systems treat recommendation as a purely preference-

matching task, often ignoring the temporal, cognitive, and emotional fluctuations that shape user behavior. In the context of reading, where attention is fragmented, emotions shift, and user intent may be ambiguous, such static models fail to generate sustained engagement. Furthermore, they struggle with the cold-start problem, content sparsity, and the need for deeper semantic comprehension of books and reviews [18].

Recent advances in deep learning and representation learning have led to significant improvements in capturing non-linear relationships and latent factors. However, even these models typically depend on extensive labeled data and remain limited in their ability to interpret context, intent, and emotion [31]. It is in this gap that generative artificial intelligence (Generative AI) has emerged as a transformative force.

Generative AI models—such as Large Language Models (LLMs) like GPT or BERT—are capable of understanding, generating, and reasoning with natural language. In the context of book recommendation, these models offer several strategic advantages. First, they enable semantic comprehension by extracting deep thematic structures, genre characteristics, and narrative tones from book descriptions and user reviews, which allows for more refined content matching [10]. Second, they support emotion-aware recommendation through the analysis of user-generated texts such as reviews or comments, allowing the system to infer sentiment and emotional preferences, thus producing recommendations that better align with the reader’s mood and psychological needs [30]. Third, they enable interactive personalization by engaging users in natural language dialogue, explaining recommendation rationales, adjusting suggestions in real time based on feedback, and even generating custom summaries or comparisons between books to support decision-making.

These capabilities position generative AI as more than a computational upgrade; it acts as a cognitive partner that guides readers through complex choices in cognitively efficient and emotionally resonant ways. In this study, we leverage generative AI not only for feature extraction but also to support reading immersion—a state that depends on both relevance and psychological readiness.

By integrating LLMs into a learning-to-rank recommendation pipeline, and combining semantic, social, and behavioral signals, this research aims to advance a new generation of recommender systems. These systems are not only technically robust, but also capable of engaging the reader as a whole person, addressing the intertwined challenges of attention fragmentation, emotional variance, and motivation loss [32].

While recent advancements in Generative AI have enabled semantic and emotional modeling, few studies have applied these capabilities to address the unique challenges of fragmented digital reading behaviors. This study aims to bridge that gap by leveraging LLMs to model cognitive variance and emotional preferences within a personalized recommendation framework.

Chapter 3. The System Framework

In this study, we propose a method for matching books and readers in a way that promotes immersive reading and provides emotionally and cognitively aligned recommendations. To support this goal, we selected Goodreads, a globally recognized book-sharing platform, as our primary data source. Goodreads offers both quantitative (e.g., ratings, votes) and qualitative (e.g., reviews) user-generated content, which allows us to understand reader behavior, preferences, and emotional engagement. Our approach—illustrated in Figure 4—follows a multi-stage pipeline designed to address the challenges of attention scarcity and fragmented digital reading habits.

To address the fragmented attention and motivational challenges observed in digital reading, this system was designed not only to predict preferences, but also to reconstruct cognitive pathways for immersive reading. As shown in Figure 4, the system incorporates semantic, emotional, and social signals through modular processing, culminating in a recommendation engine guided by generative artificial intelligence.

Firstly, we conducted data collection by retrieving the top 30 ranked books from Goodreads for each year between 1924 and 2024. To facilitate meaningful comparison and training, we curated three distinct datasets that categorize books by ranking position, genre, and publication year. Additionally, we collected metadata and review content for the first 60 representative users.

Secondly, we focused on contextual signal extraction. From the collected datasets, we derived multi-dimensional signals relevant to immersive reading. These include book-level attributes such as format (e-book/audiobook), genre, and award status, as well as reader-generated information including ratings, reviews, review sentiment, and our proposed Reading Engagement Index (REI).

Thirdly, we designed a matching engine using the extracted features. This engine integrates large language models (LLMs) to analyze semantic relationships and emotional tone. It ensures

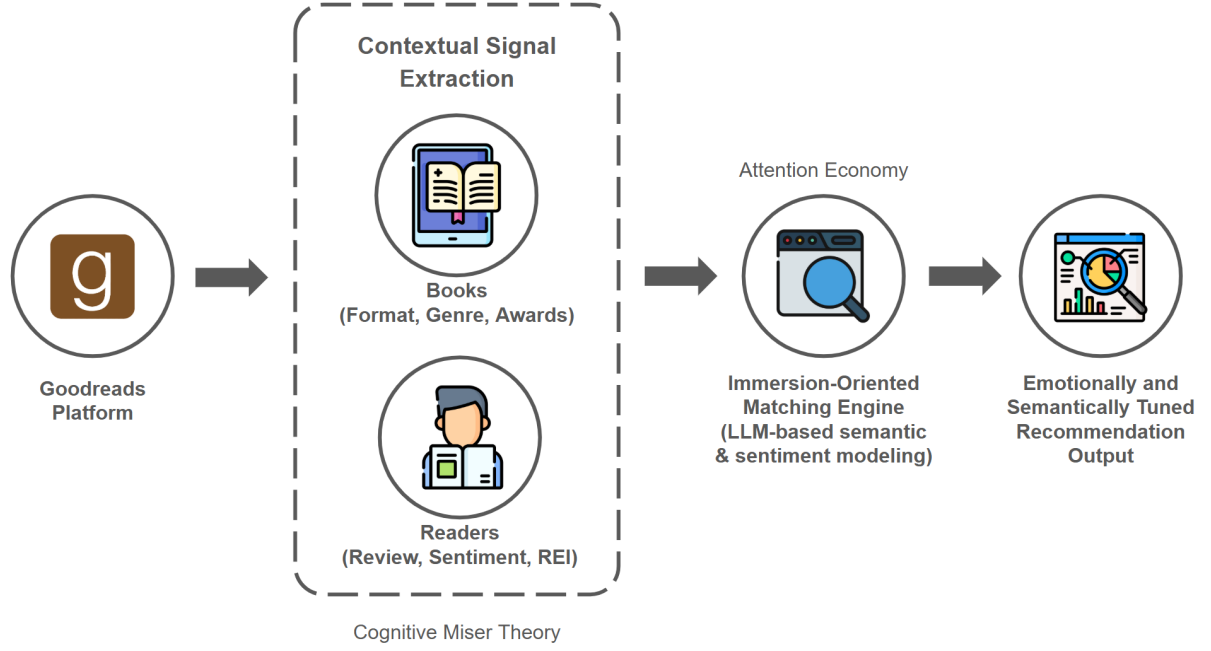


Figure 4: Immersive Book Recommendation Workflow: From Contextual Signal Extraction to Emotionally Aligned Output

that recommended books align with both the reader’s preferences and their cognitive/emotional state, thereby promoting deeper reading engagement.

Finally, we carried out recommendation generation by applying multiple learning-to-rank algorithms—such as XGBoost, SVM, and Random Forest—to generate a personalized list of recommended books. The results were further filtered and tuned to emphasize emotional compatibility, accessibility, and relevance, ultimately optimizing the reading list for immersive experience.

The proposed architecture consists of four core modules, each contributing a different layer of analysis—from raw data processing to immersive reading prediction. The data construction module is responsible for preparing structured datasets that represent both books and readers in a usable format. The book immersion feature module focuses on extracting key attributes related to format availability, reader engagement, and social visibility. Complementing this, the cognitive-emotional modeling module leverages Large Language Models (LLMs) to capture sentiment polarity and behavioral variance from reader reviews and rating patterns. Finally, the LLM-assisted matching engine ranks books by emphasizing semantic resonance and emotional alignment, ensuring that the recommendation output is both contextually meaningful and affectively engaging.

(1) **Data Construction Module:** This module is responsible for the procedures of data collec-

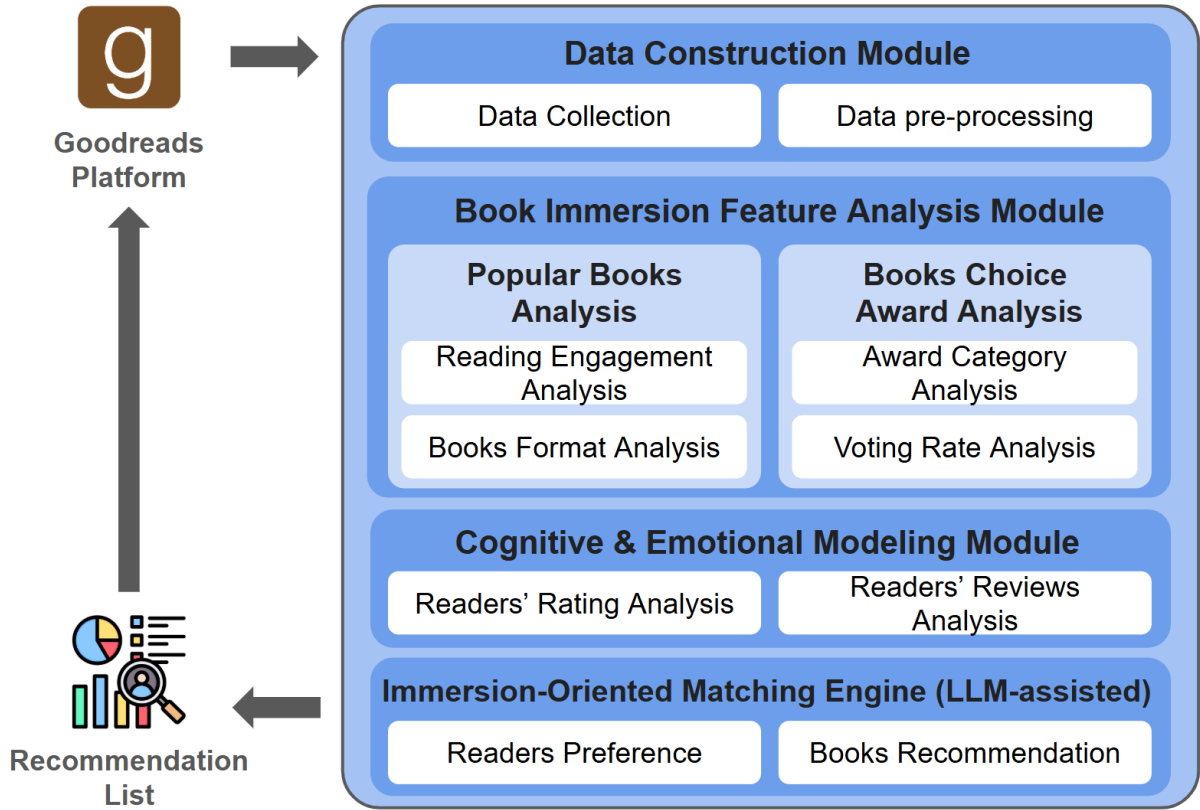


Figure 5: The System Framework of the Immersive Reading Recommendation Model

tion and pre-processing. First, we obtained data from the Goodreads platform using web crawling techniques, focusing on top-ranked books across years. The raw data includes metadata, reader interactions, reviews, and voting records. Next, we conducted data pre-processing to clean and transform the dataset into a structured format, ensuring quality and compatibility for downstream analytical tasks. This two-step process forms the foundation for subsequent signal extraction and modeling.

- (2) **Book Immersion Feature Analysis Module:** This module focuses on extracting book-level features that are relevant to immersive reading. It consists of two subcomponents: Popular Books Analysis and Books Choice Award Analysis. The first subcomponent analyzes engagement-related characteristics, including book formats (e.g., hardcover, e-book, audiobook) and reader volume (current vs. intended readers), forming the basis of the Digital Format Index (DFI) and Reading Engagement Index (REI). The second subcomponent focuses on recognition-related metrics, such as award categories and voting statistics, to capture social visibility and perceived value. These features collectively help us assess the immersive potential of each book.

- (3) **Cognitive & Emotional Modeling Module:** This module analyzes user-generated behavioral signals, including rating distributions and review text. The Readers' Rating Analysis extracts cognitive response patterns after reading, while the Readers' Reviews Analysis leverages large language models (LLMs) to perform semantic and sentiment analysis on textual feedback. This module bridges traditional behavioral data with inferred psychological engagement, enabling a deeper understanding of user preference and attention flow.
- (4) **Immersion-Oriented Matching Engine (LLM-assisted):** This module integrates the contextual features derived from the previous modules and applies multiple learning-to-rank algorithms (e.g., XGBoost, SVM, Random Forest). With the support of LLM-based embeddings and sentiment vectors, the engine produces personalized recommendations that **prioritize emotional resonance, semantic relevance, and format accessibility. The goal is not merely to predict high scores but to match readers with books that align with their cognitive effort profiles and encourage immersive reading behavior.**

3.1 Data Construction Module

This module is responsible for constructing structured datasets from Goodreads. Web scraping techniques were used to collect metadata for top-ranked books from 1924 to 2024, including reader reviews, ratings, votes, award tags, and reading status. The data were cleaned and organized into separate datasets by **ranking position, genre, and year**. Additionally, data from 60 active readers were sampled to support user profiling. This module enables the extraction of features used in engagement modeling, sentiment analysis, and reader-book matching.

3.1.1 Data Collection

Goodreads is an American social cataloging website and a subsidiary of Amazon that allows people to search its database of books, annotations, quotes, and reviews. Therefore, it serves as a suitable data source to gain a lot of information about readers and books around the world. Three datasets, namely *the popular books dataset*, *the best books dataset* and *the readers' reviews dataset*, were obtained using a Python crawler.

- (1) **The Popular Books Dataset:** This dataset spans 101 years, from 1924 to 2024, and consists of the top 200 books most frequently added to Goodreads members' shelves each year.

We have selected the top 30 books for analysis. Each book entry contains the following attributes: format, number of current readers, number of intending readers, average rating score, total number of ratings, and number of reviews.

- (2) **The Best Books Dataset:** This dataset is different from the one mentioned above. It started in 2011, where at the end of each year readers are invited to vote for what they consider to be the best books of the year. Taking the voting rules for 2024 as an example, there are 20 nominated books in each category, with a total of 15 categories. In the first round of voting, readers can select one book in each category, and in the second round, they narrow it down to the top 10 books in each category. This dataset includes the total number of votes for the year, the total number of votes for each category, the number of votes on the books that ranks highest in each category, and the number of votes and related book information for the 20 books nominated in each category.
- (3) **The Readers' Reviews Dataset:** This dataset is about the top 60 reviewers rate book scores and reviews given to books in the world. This data set will include the reviewer's name, the books' title be reviewed, the score the reviewer gave to the books, and the books' reading experience (review).

3.1.2 Data Pre-processing

After gathering all the necessary data, we performed data preprocessing to ensure its quality and interpretability. As part of this process, we verified that the data was up-to-date. Two tasks were performed to achieve the objectives described in the following.

First, on Goodreads, the number of readers currently reading the book was shown in the form of '123k', '1m', etc. For convenience in our data analysis, we converted '123k' to 123000 and '1m' to 1000000.

Second, in all numerical values on Goodreads, we observed that, for ease of readability, numbers appear in the format '2,902,399', where a comma separates every three digits. For our analysis, we converted these numbers into pure numeric form by removing all commas. For example, '2,902,399' was converted to '2902399'.

3.2 Book Immersion Feature Analysis Module

This module captures book-level features relevant to immersion. It comprises two subcomponents. Popular Books Analysis derives two indicators: the Digital Format Index (DFI), based on the availability of e-book and audiobook formats; and the Reading Engagement Index (REI), computed from Goodreads' "currently-reading" and "to-read" counts. These metrics reflect accessibility and behavioral interest. Books Choice Award Analysis analyzes user votes and award types, measuring perceived recognition and social validation. Together, these features provide insight into each book's public resonance and immersive potential.

3.2.1 Popular Books Analysis

We will use the collected *The Popular Books Dataset* to observe the first aspect: *Reading Engagement Analysis* to analyze current and potential readers of each book. In addition, we can also observe the popularity of each book and its different formats, such as the difference between traditional printed books and e-books or audiobooks. This is the second aspect of observation: *Books Format Analysis*.

Reading Engagement Analysis

In this analysis, a deep understanding of the demographics of the reader is crucial. Segmenting readers into two main categories: those currently reading, and those intending to read but have not started yet, can provide valuable insights for us regarding the popularity of books and potential reading demands. This classification method helps to understand the level of interest readers have in specific books and identifies which books have the potential to translate into actual reads. The following are the definitions of reader categories:

Current Readers (CR): This category includes people who are currently reading the book, which reflects current reading activity and engagement with the book.

Intending Readers (IR): This category refers to those who have expressed interest in the book and plan to read it at some point in the future but have not yet started. This indicator helps to gauge the potential reading market for the book.

We can design the following three formulas to calculate these categories' ratios:

$$ActiveRatio_i(AR_i) = \frac{CR_i}{CR_i + IR_i} \quad (3.1)$$

$$PotentialRatio_i(PR_i) = \frac{IR_i}{CR_i + IR_i} \quad (3.2)$$

Furthermore, to quantify the general engagement and interest of the readers in these different categories for a specific book i , we can design an index for each book.

$$ReadingEngagementIndex_i(REI_i) = AR_i \times \alpha + PR_i \times \beta \quad (3.3)$$

where $\alpha + \beta = 1$ and $\alpha, \beta \geq 0$

α and β represent the weights assigned to these two categories, reflecting their relative importance in measuring the popularity of a book and its potential reading interest. We define α and β as 0.75 and 0.25 respectively, as readers currently engaged in reading directly reflect current reading participation and activity.

Through these formulas, we obtain a numerical indicator that comprehensively considers the number of different types of readers and the relative importance of categories, providing a comprehensive assessment of the current and potential reading engagement of a book. This approach helps us know about each book's reading engagement.

Books Format Analysis

In recent years, with the advancement of technology and digital transformation, the popularity of e-books and audiobooks in the global reading market has increased dramatically. This shift has not only influenced people's reading habits, but has also reshaped the business models of the publishing industry. The popularity of e-books and audiobooks is attributed to several key factors: portability, accessibility, and personalized options. These formats enable readers to access a wide range of books anytime, anywhere, and adjust their reading experience according to personal preferences, such as adjusting font size or listening speed.

Every book i has many formats, such as hardcover, e-book, audiobook, paperback, etc. Each format would have many editions. Therefore, we can design an index for each book.

$$DigitalFormatIndex_i(DFI_i) = \frac{N_{EbookEdition_i} + N_{AudiobookEdition_i}}{N_{TotalEdition_i}} \quad (3.4)$$

Here, $N_{EbookEdition}$ refers to the number of editions for the e-book, $N_{AudiobookEdition}$ refers to the

number of editions for the audiobook, and $N_{TotalEdition}$ refers to the total number of editions for this book.

3.2.2 Books Choice Award Analysis

We will use the collected *The Bests Books Dataset* to observe the first aspect: *Award Category Analysis* to analyze the changes in award categories over the past 14 years. In addition, we can also observe the voting rate of each nominated book. This is the second aspect of observation: *Voting Rate Analysis*.

Award Category Analysis

From 2011 to last year 2024, there were equal to or more than 15 categories each year. The variety is similar each year but not exactly the same. We will analyze the categories and vote counts for each year to see which categories are the most popular. Some categories, such as fiction, have been on the list for 14 consecutive years. Some have created new categories over time, such as audiobooks.

We define a binary matrix M where each row represents a year and each column represents a category. The value in the matrix is defined as:

$$M_{ij} = \begin{cases} 1, & \text{if category } c_j \text{ exists in year } y_i \\ 0, & \text{if category } c_j \text{ does not exist in year } y_i \end{cases}$$

where:

- M is an $(n \times m)$ matrix.
- n is the number of years (2011 to 2024, so $n = 14$).
- m is the total number of unique categories.

To determine how many years a particular category c_j has been present:

$$S_j = \sum_{i=1}^n M_{ij}$$

where S_j represents the total number of years in which category c_j appeared.

Voting Rate Analysis

In the previous section, we examined the presence of different award categories across the years to understand which categories remained stable or varied. In this section, we analyze the voting proportions for each category to evaluate their popularity. Specifically, we focus on two key metrics:

The proportion of votes that each category receives out of the total votes in a given year is calculated as follows:

$$V_j = \frac{T_j}{\sum_{k=1}^m T_k} \quad (3.5)$$

where:

- V_j represents the voting share of category c_j in a given year.
- T_j is the total number of votes received by category c_j .
- m is the total number of award categories in that year.

This metric helps compare the popularity of different categories and observe trends over time.

The proportion of votes received by the top-ranking work within each category, relative to the total votes in that category, is calculated as:

$$W_j = \frac{T_{j,\max}}{T_j} \quad (3.6)$$

where:

- W_j represents the winning vote share for category c_j .
- $T_{j,\max}$ is the highest number of votes received by a single work within category c_j .
- T_j is the total number of votes in category c_j .

This metric evaluates whether a category is dominated by a single work or if competition is more evenly distributed. If W_j approaches 1, it indicates that the winning work has a highly concentrated share of votes. Conversely, a lower value suggests a more competitive category.

By analyzing these two metrics, we can identify voting trends across different categories, assess the intensity of competition, and determine which categories maintain stable popularity over time.

3.3 Cognitive & Emotional Modeling Module

This module models reader psychology by analyzing behavioral and emotional signals. It consists of: **Readers' Rating Analysis** evaluates post-reading behavior by measuring rating distributions. High variance may indicate ambiguity or emotional polarization, while consistency suggests stable cognitive alignment. **Readers' Reviews Analysis** applies LLMs to reviews to extract semantic representations and classify sentiment polarity. These insights help form reader emotion profiles, enabling personalized matching that adapts to affective needs.

By combining both analytical and inferential approaches, this module constructs a nuanced understanding of the reader's cognitive disposition and emotional state.

3.3.1 Readers' Rating Analysis

In this section, we analyze the rating behavior of 60 selected reviewers, each of whom has reviewed 100 different books. We examine how each reviewer's ratings deviate from the Goodreads average rating for the books they reviewed.

For each reviewer j , let B_j denote the set of 100 books they reviewed. The standard deviation of their ratings compared to the Goodreads average rating is computed as follows:

$$\sigma_j = \sqrt{\frac{1}{|B_j|} \sum_{i \in B_j} (R_{ij} - \bar{R}_i)^2} \quad (3.7)$$

where:

- R_{ij} is the rating given by reviewer j to book i .
- $\bar{R}_i$ is the average rating of book i from all Goodreads users.
- B_j represents the set of books reviewed by reviewer j , with $|B_j| = 100$.

A higher σ_j indicates greater deviation from the general rating consensus, whereas a lower σ_j suggests that the reviewer's ratings closely align with the average ratings.

The overall variability among all reviewers is computed as:

$$\sigma_{overall} = \frac{1}{n} \sum_{j=1}^n \sigma_j \quad (3.8)$$

where $n = 60$ is the total number of reviewers.

To determine whether a reviewer systematically gives higher or lower ratings compared to the Goodreads average, we compute the reviewer's average rating deviation:

$$D_j = \frac{1}{|B_j|} \sum_{i \in B_j} (R_{ij} - \bar{R}_i) \quad (3.9)$$

where:

- $D_j > 0$ indicates that reviewer j tends to give higher ratings than the general Goodreads audience.
- $D_j < 0$ indicates that reviewer j tends to give lower ratings than the general Goodreads audience.
- $D_j = 0$ means that the reviewer's ratings align exactly with the Goodreads average.

The overall rating bias across all 60 reviewers is given by:

$$D_{overall} = \frac{1}{n} \sum_{j=1}^n D_j \quad (3.10)$$

A positive $D_{overall}$ suggests that this group of reviewers tends to rate books higher than the Goodreads average, while a negative $D_{overall}$ suggests a tendency toward lower ratings.

These findings contribute to understanding rating consistency and can be used to refine book recommendation algorithms.

3.3.2 Readers' Reviews Analysis

To better understand the emotional tendencies of individual reviewers, we conduct sentiment analysis on the review content written by the top 60 reviewers. Each reviewer has contributed 100 reviews, each corresponding to a different book. Using a Large Language Model (LLM)-based sentiment classifier, each review is categorized as either positive ($S = 1$), neutral ($S = 0$), or negative ($S = -1$).

For each reviewer j , we compute the average sentiment score across all their reviews:

$$\bar{S}_j = \frac{1}{|R_j|} \sum_{i \in R_j} S_{ij} \quad (3.11)$$

where:

- S_{ij} is the sentiment score of reviewer j for book i .
- R_j is the set of books reviewed by reviewer j , with $|R_j| = 100$.

This score $\bar{S}_j$ ranges from -1 to 1, where values closer to 1 indicate a more positive emotional tone across all reviews, and values closer to -1 suggest a more negative tone.

To further analyze the sentiment composition, we calculate the proportion of each sentiment category for every reviewer:

$$P_{pos}^{(j)} = \frac{\sum_{i \in R_j} \mathbb{1}(S_{ij} = 1)}{|R_j|}, \quad P_{neu}^{(j)} = \frac{\sum_{i \in R_j} \mathbb{1}(S_{ij} = 0)}{|R_j|}, \quad P_{neg}^{(j)} = \frac{\sum_{i \in R_j} \mathbb{1}(S_{ij} = -1)}{|R_j|} \quad (3.12)$$

These proportions reveal whether a reviewer tends to express more favorable, neutral, or critical opinions in their written reviews.

This analysis complements the numerical rating analysis by capturing emotional nuances that are not reflected in numeric scores alone.

3.4 Immersion-Oriented Matching Engine (LLM-assisted)

The final module integrates the contextual signals extracted from books and readers to generate personalized recommendations. Feature vectors—including REI, DFI, award signals, semantic embeddings, and sentiment scores—are input into a learning-to-rank pipeline.

Three ranking strategies were implemented: point-wise, pair-wise, and list-wise. Algorithms such as XGBoost, Random Forest, and Support Vector Machines (SVM) were tested and compared. The engine also benefits from LLM-derived embeddings, which enrich feature representations by capturing latent thematic and emotional dimensions. The result is a ranked book list optimized not only for relevance, but for cognitive effort alignment, emotional resonance, and immersive continuity.

Figure 5 visualizes the architecture that underpins these modules, emphasizing the flow from contextual data extraction to human-centered recommendation output.

3.4.1 Readers' Preference

In this study, we construct reader preference models based solely on the rating scores that each reader has assigned to different books. This focus on explicit feedback enables a direct

mapping between reader preferences and book attributes. These ratings serve as direct indicators of personal interest and are treated as the primary input features for our recommendation models.

For each reader-book pair, we collect the actual rating value given by the reader and use this as training data. Under the point-wise framework, the model learns to predict an absolute preference score for each individual book. In the pair-wise setting, the model compares two books and determines which one is more preferred by the reader. For the list-wise approach, the entire ranked list of books is considered during training to optimize the global ranking quality.

This design enables the model to learn the relationship between rating scores and reader preferences, and to build personalized recommendation logic accordingly.

3.4.2 Books Recommendation

Once the preference models are trained, the recommendation process is conducted based on the predicted preference scores for each reader-book pair. The system ranks all candidate books in descending order of their predicted scores, resulting in a tailored recommendation list for each individual reader.

The recommendation process includes the following steps: **Score Prediction:** For every reader-book pair, the trained model predicts a preference score. **Ranking:** All books are sorted by their predicted scores from highest to lowest, indicating the level of interest for each reader. **Recommendation List Generation:** The top-ranked items are selected to form the final recommendation list.

This module is designed with flexibility and modularity in mind. This flexibility ensures future scalability for integrating additional features such as user demographics or advanced semantic analysis. It allows for future integration of additional book features or reader information to further enhance the system's adaptability and recommendation quality.

Chapter 4. Experiments

In this chapter, we provide a detailed account of the experimental process, including the formulas applied in each module, illustrative examples, and explanations of how we utilized the collected dataset. We also present key insights derived from our analysis. Our goal is to offer a comprehensive overview of our methodology, demonstrating the application of formulas and datasets to effectively convey our findings.

4.1 Data Construction Module

4.1.1 Data Collection

As described in Chapter 3, we will utilize Goodreads as our data source and Python crawler technology to collect books' and readers' information. This experiment will involve collecting data from popular books' information. The first dataset will include the top 30 popular books over a total of 101 years of data from January 1, 1924, to December 31, 2024. Each book includes the following features: the book's format, the number of readers currently reading the book, the number of readers who want to read the book, the average rating score, the total number of ratings, and the number of reviews. We have collected information on 3030 books in total. Then, the second dataset will include the 20 nominated books in each of the 15 categories since 2011. This dataset contains the total number of votes for the year, the vote count for each category, the number of votes received by the top-ranked book in each category, and the vote counts along with related information for all 20 nominated books in every category. Finally, we also collected reader data. This dataset contains the reviewer's name, the score the reviewer gave the book, and the review given to the book. We collected book reviews from the world's top 60 readers and obtained a total of 6000 ratings and 6000 reviews.

4.1.2 Data pre-processing

The dataset exists in various forms and requires pre-processing prior to analysis. The pre-processing steps are as follows.

The Popular Books Dataset and The Best Books Dataset: When collecting information from the books, the number of readers currently reading the book, the number of readers who want to read the book, the total number of ratings, and the number of reviews are presented in a simplified manner. Therefore, before performing the experiment, it needs to be converted to numerical values. Furthermore, in all numerical values on Goodreads, we speculate that, for ease of readability, the format appears as '2,902,399', where a comma separates every three digits. For convenience in our data analysis, we will convert these numbers into numeric form and remove all commas. For example, '2,902,399' will be converted to '2902399'.

Before	After
123k	123000
1m	1000000
2,902,399	2902399

Table 1: The Example of Converting Number of Books

4.2 Book Immersion Feature Analysis Module

To better understand the current and potential engagement levels of readers with various books, we conducted the following analyses. Books are at the core of our analysis. By examining the information within these books, we can determine their current popularity, their potential readership, and the likelihood of attracting future readers. This forms the first area of our study: *Reading Engagement Analysis*. Additionally, we assess the popularity of each book across different formats, such as traditional printed books, e-books, and audiobooks. This constitutes the second area of our observation: *Books Format Analysis*.

4.2.1 Reading Engagement Analysis

Based on the above description, in this analysis, we calculate the *ActiveRatio (AR)* and *PotentialRatio (PR)* for each book. As shown in Figure 6, we can observe that the AR and PR

of top books in the past decade. An example of the calculation for these ratios is shown in Table 2.

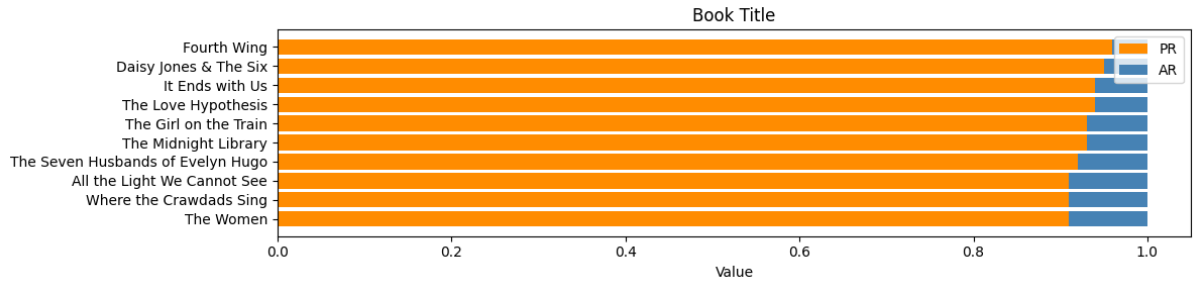


Figure 6: The Top Books of The Past Decade: ActiveRatio and PotentialRatio

It can be observed that books with a higher ActiveRatio are likely to maintain stronger current readership engagement.

First, we obtain the *CurrentReaders* (CR) and *IntendingReaders* (IR) of a book. By adding these two values together, we get the total number of readers for that book. The *ActiveRatio* (AR) is calculated by dividing the number of *CurrentReaders* (CR) by the total number of readers, while the *PotentialRatio* (PR) is obtained by dividing the number of *IntendingReaders* (IR) by the total number of readers.

Finally, we multiply the *ActiveRatio* (AR) by alpha (α) and the *PotentialRatio* (PR) by beta (β). The sum of these two weighted values gives us the *ReadingEngagementIndex* (REI) for each book.

Title	CurrentReaders (CR)	IntendingReaders (IR)	ReadingEngagementIndex (REI)
The Women	76900	83355	0.48993
The Maid	668	29632	0.26102

Table 2: The Example of Calculating ReadingEngagementIndex (REI)

4.2.2 Books Format Analysis

In addition to reader engagement, examining the available book formats can reveal important trends in digital consumption. In this analysis, we calculate the *DigitalFormatIndex* (DFI)

for each book. In figure 7, it shows the DFI of the most popular book every decade from 2024 to 1934. Table 3 provides an example of how the index is calculated.

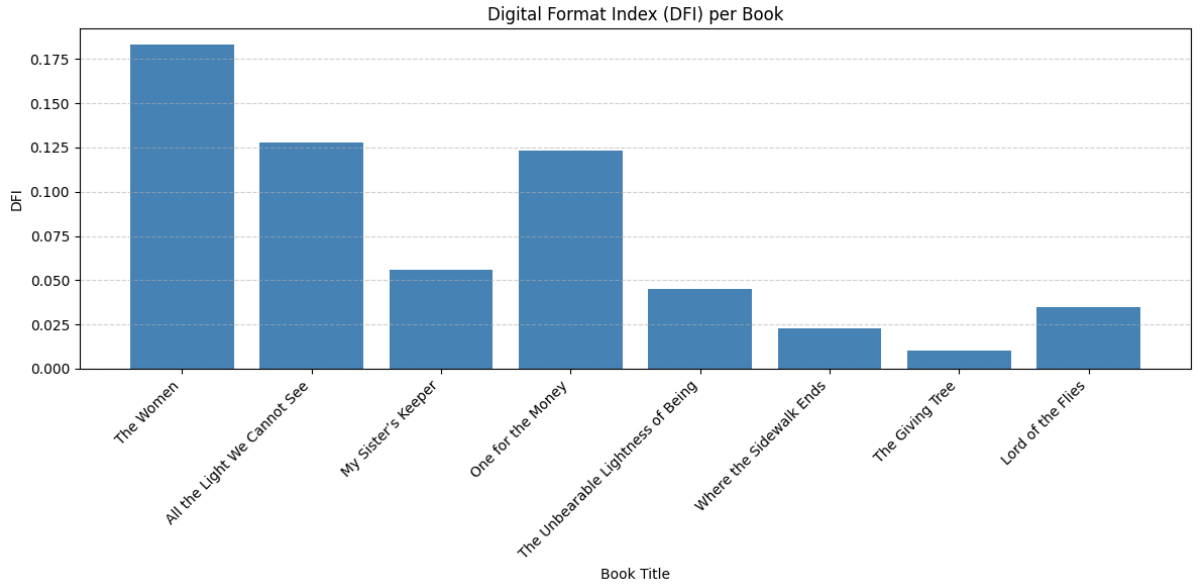


Figure 7: The DFI of The Most Popular Book for Each Decade from 1954 to 2024

The trend shows an increasing preference for digital formats, especially in recent decades.

First, we obtain the $N_{EbookEdition}$ and $N_{AudiobookEdition}$ of a book. By adding these two values together, we get the total number of digital formats (e-book editions + audiobook editions) for that book. Then, we also obtain the $N_{TotalbookEdition}$ of the book. Finally, the *DigitalFormatIndex* (DFI) is calculated by dividing the total number of digital editions by the total number of editions.

Title	$N_{TotalbookEdition}$	$N_{EbookEdition}$	$N_{AudiobookEdition}$	DigitalFormatIndex (DFI)
The Women	71	9	4	0.1831
The Maid	76	2	6	0.10526

Table 3: The Example of Calculating DigitalFormatIndex (DFI)

4.2.3 Award Category Analysis

From 2011 to 2024, a total of 27 categories have appeared in the Goodreads Choice Awards, reflecting the evolving interests and reading preferences of the public over time. Among these,

well-established categories such as *fiction*, *historical fiction*, and *mystery-thriller* have consistently been featured throughout the past 14 years. Additionally, some categories have undergone modifications or adaptations to better align with shifting literary trends. It is also worth noting that the *audiobook* category was introduced as a new addition in 2024, marking an expansion in recognition of the growing influence of digital and audio-based reading experiences. This inclusion highlights the increasing significance of audiobooks in contemporary literary consumption. Figure 8 and table 4 provides a detailed statistical overview of the various categories that have been part of the Goodreads Choice Awards, offering insights into the diversity and changes within the awards over the years.

Category	Frequency (years)
fiction	14
historical-fiction	14
mystery-thriller	14
romance	14
fantasy	14
science-fiction	14
horror	14
young-adult-fantasy	14
young-adult-fiction	14
nonfiction	14
memoir-autobiography	14
history-biography	14
humor	13
middle-grade-children's	12
graphic-novels-comics	12
poetry	12
debut-novel	12
picture-books	10
food-cookbooks	10
science-technology	6
paranormal-fantasy	3
romantsay	2
goodreads-author	2
audiobook	1
best-of-the-best	1
business-books	1
travel-outdoors	1

Table 4: Summary of Category Appearance Frequency from 2011 to 2024

The dominance of traditional genres such as fiction and historical fiction highlights consistent reader interest over time.

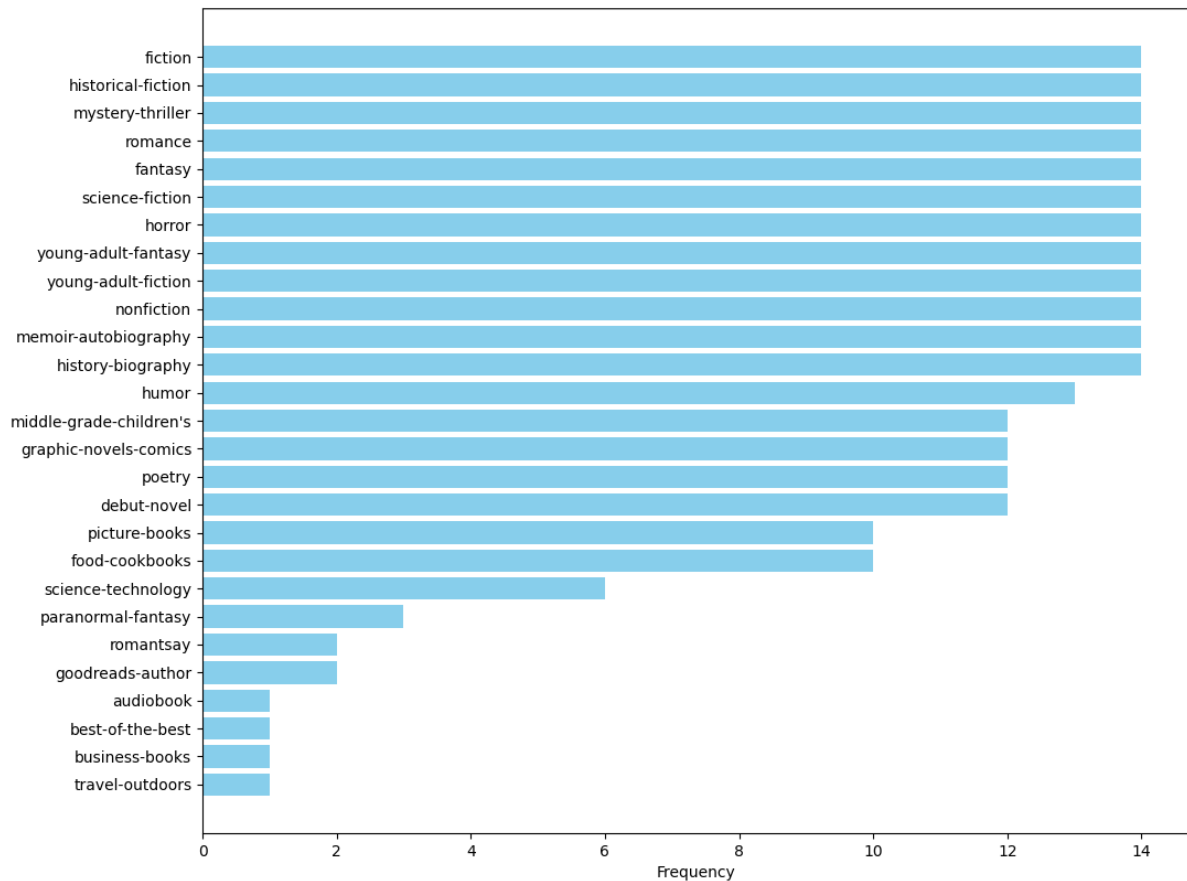


Figure 8: The Statistic of the Goodreads Choice Award Categories

These results provide an overview of how category diversity has evolved over time in the Goodreads Choice Awards.

4.2.4 Voting Rate Analysis

Over the past 14 years, we calculated the vote share within each category and analyzed 12 consistently appearing categories. Our findings indicate that *fiction*, *young-adult fiction*, and *nonfiction* categories generally exhibit a more diverse competitive landscape, as the highest vote share for top-ranking books in these categories does not show excessive dominance in most years. This suggests that these categories feature a larger number of popular books, leading to a more distributed voting pattern and greater variety in reader preferences.

On the other hand, some categories demonstrate a more concentrated voting trend, such as *historical fiction*, *mystery-thriller*, and *memoir & autobiography*. In specific years, the highest vote share within these categories is significantly higher than in other categories, indicating a higher degree of market dominance. This could suggest that readers in these categories exhibit strong loyalty to particular books, or that a lack of strong competitors in certain years allows

a single book to win by an overwhelming margin. For example, in 2015, the *mystery-thriller* category recorded a notably high highest vote share compared to other categories, likely due to the presence of an exceptionally popular book that attracted a large number of votes, thereby reducing overall competition within the category. This highlights that different categories exhibit varying levels of competitiveness across years.

Year	Fiction	Historical-Fiction	Mystery-Thriller	Romance	Fantasy	Science-Fiction	Horror	Young-Adult-Fantasy	Young-Adult-Fiction	Nonfiction	Memoir-Autobiography	History-Biography
2011	0.1321	0.1365	0.1393	0.1641	0.2313	0.1847	0.1722	0.1821	0.1362	0.2324	0.1985	0.2819
2012	0.1549	0.1263	0.3513	0.2579	0.1726	0.1797	0.1804	0.2420	0.4288	0.2046	0.1948	0.1916
2013	0.1797	0.1873	0.1950	0.1770	0.3652	0.2179	0.3322	0.2085	0.1860	0.1411	0.3783	0.2609
2014	0.1965	0.2070	0.1674	0.2128	0.2203	0.2088	0.1600	0.1978	0.1861	0.1468	0.1893	0.1865
2015	0.1630	0.3173	0.3955	0.1687	0.1705	0.2262	0.1749	0.1404	0.1763	0.1545	0.1321	0.2015
2016	0.1268	0.1339	0.1779	0.2048	0.4055	0.2465	0.1735	0.1917	0.1543	0.3546	0.1636	0.1384
2017	0.1441	0.1306	0.1796	0.1542	0.2647	0.1941	0.3230	0.2461	0.2320	0.1467	0.1981	0.1745
2018	0.1665	0.2239	0.1892	0.1728	0.2516	0.2265	0.2638	0.2228	0.1197	0.1865	0.2660	0.2898
2019	0.2844	0.2464	0.2115	0.2374	0.1610	0.1832	0.4039	0.1624	0.1459	0.1254	0.1459	0.1996
2020	0.1437	0.2695	0.1981	0.1810	0.2246	0.1934	0.3185	0.1715	0.1383	0.1577	0.2640	0.2214
2021	0.1797	0.2576	0.1496	0.2036	0.2850	0.3297	0.2290	0.1537	0.1210	0.1857	0.2361	0.1178
2022	0.1853	0.2234	0.1749	0.2851	0.2126	0.1971	0.1498	0.1334	0.2047	0.1478	0.5410	0.1302
2023	0.3415	0.1594	0.1664	0.2442	0.1662	0.1878	0.2643	0.2935	0.1851	0.1157	0.3412	0.1849
2024	0.1787	0.4636	0.1445	0.2382	0.1688	0.2829	0.2115	0.2186	0.2073	0.1481	0.2677	0.1604

Table 5: Voting Share of Major Categories from 2011 to 2024

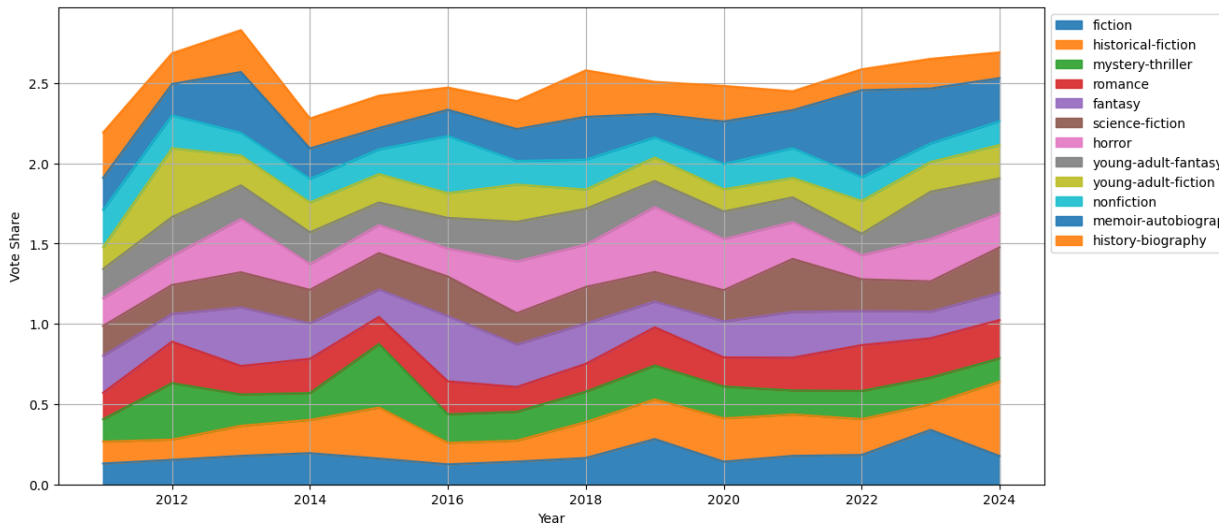


Figure 9: The Winning Vote Share by Category from 2011 to 2024

The following analysis focuses on the concentration of votes within each category, providing insights into the competitiveness of awards.

4.3 Cognitive & Emotional Modeling Module

Readers play a crucial role as well. By examining the ratings and reviews from top reviewers, we can gain a clearer understanding of each book’s content. In this module, we can conduct two types of analyses: Readers’ Rating Analysis and Readers’ Reviews Analysis.

4.3.1 Readers' Rating Analysis

In total, we analyzed the rating patterns of 60 reviewers, each of whom rated 100 different books. The average standard deviation across all reviewers was calculated as 0.9153, with the maximum at 2.0206 and the minimum at 0.4894. This suggests that while most reviewers show moderate consistency with public opinion, a few demonstrate highly variable rating behaviors.

Reviewer	Standard Deviation (σ_j)
Jamie	2.0206
Soobie	1.6608
Darcy	1.5978
Theediscerning	1.5349
Sarah	1.3879

Table 6: Top 5 Reviewers by Rating Standard Deviation (σ_j)

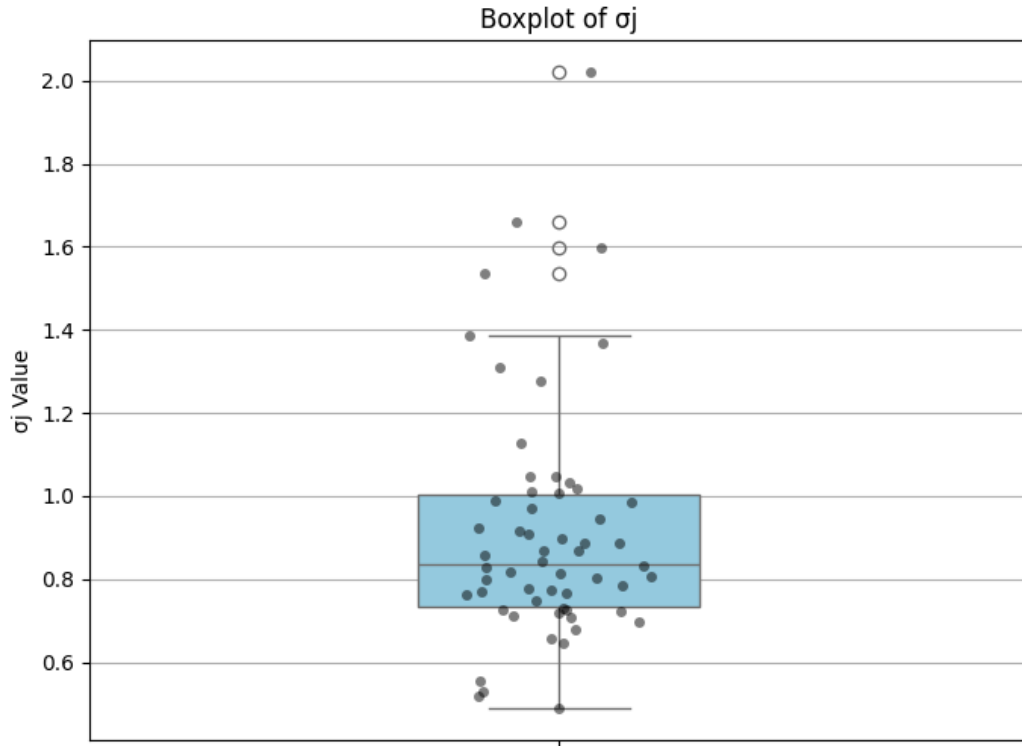


Figure 10: The Box Plot of Standard Deviation of Individual Reviewers

Furthermore, we computed the average rating deviation for each reviewer compared to the

Goodreads average ratings. The mean deviation was 0.2459, indicating that, on average, reviewers tend to give slightly higher scores than the Goodreads platform average. The maximum positive deviation reached 0.9495, while the minimum was -1.2609, highlighting significant differences in individual rating tendencies.

Reviewer	Average Deviation (D_j)
Louisa	0.9495
Kathleen	0.8328
Raecharmed	0.8326
Kaye	0.8310
Theresa	0.8258

Table 7: Top 5 Reviewers by Average Rating Deviation (D_j)

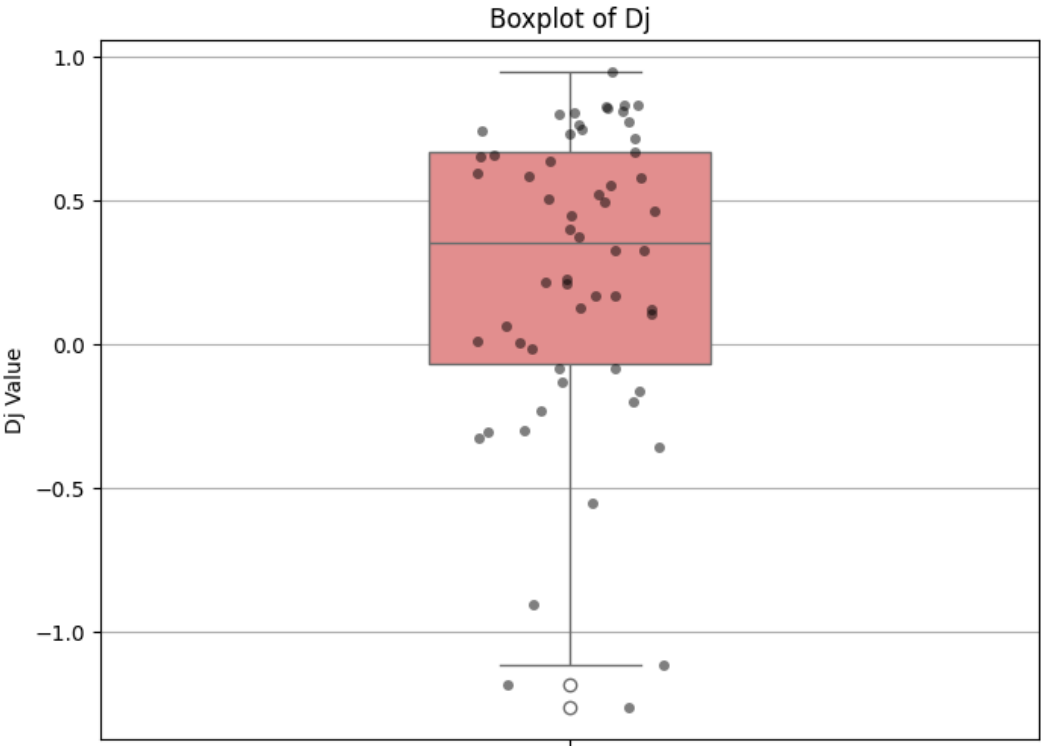


Figure 11: The Box Plot of Average Rating Deviation of Individual Reviewers

This reveals that while most reviewers exhibit moderate variability, a few display significantly divergent rating behaviors.

These statistical findings highlight the variability and bias patterns among individual reviewers.

4.3.2 Readers' Reviews Analysis

To gain deeper insights into the emotional patterns of individual reviewers, we performed sentiment analysis on the written reviews from the top 60 reviewers. Each of these reviewers wrote 100 reviews, with each review pertaining to a unique book. We employed a sentiment classification system powered by a Large Language Model (LLM) to label each review as positive ($S = 1$), neutral ($S = 0$), or negative ($S = -1$).

The resulting score $\bar{S}_j$ captures the overall emotional tone in the reviewer's writing, with values closer to 1 indicating more positive reviews and values closer to -1 indicating more negative ones.

Figure 12 illustrates the sentiment mean of each reviewer. Most reviewers exhibit high sentiment means, indicating that their reviews tend to be positive. However, a few reviewers, such as Trigger and Jamie, exhibit more negative sentiment tones, and their average sentiment scores fall below zero.

Reviewer	Average Sentiment Score
Wendy	1.00
Kathleen	1.00
Jennifer	1.00
Jeanne	1.00
Brenda	1.00

Table 8: Top 5 Reviewers by Sentiment Score

Reviewer	Average Sentiment Score
Trigger	-0.71
Jamie	-0.27
Theediscerning	-0.25
Denise	0.00
Soobie	0.03

Table 9: Top 5 Reviewers by Negative Sentiment Score

Overall, most reviewers exhibit positive sentiment trends, with a few showing notably negative tendencies. The sentiment analysis results further complement the rating behavior analysis, revealing the emotional tendencies of reviewers.

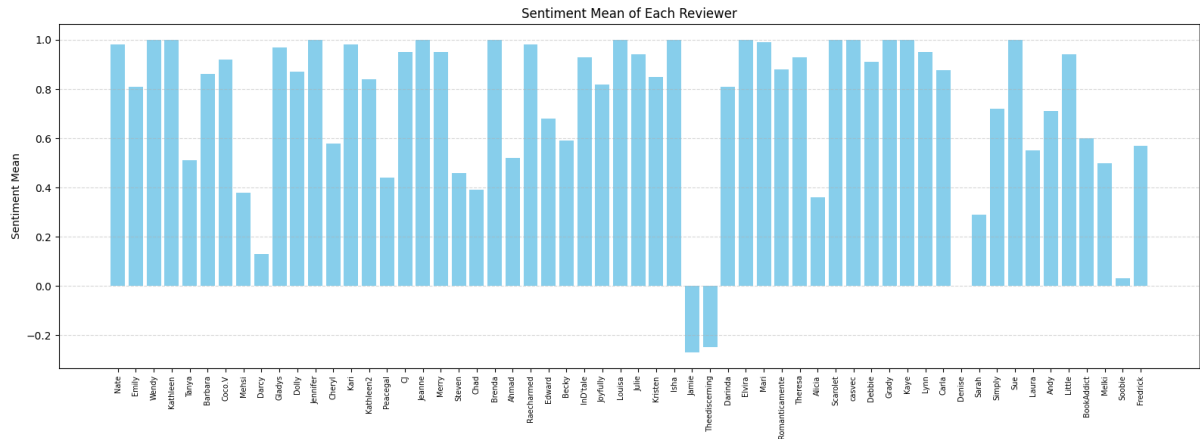


Figure 12: Sentiment Mean of Each Reviewer

To examine whether reviewers who express more positive or negative sentiments also rate books higher or lower compared to the Goodreads average, we plotted the sentiment mean $\bar{S}_j$ of each reviewer against their average rating deviation D_j .

Figure 13 presents the scatter plot of sentiment mean vs. rating deviation. The plot reveals a general trend: reviewers with higher sentiment means tend to give higher ratings than the average. Conversely, reviewers with lower sentiment means often exhibit more negative rating deviations, indicating that their scores are generally below the Goodreads average.

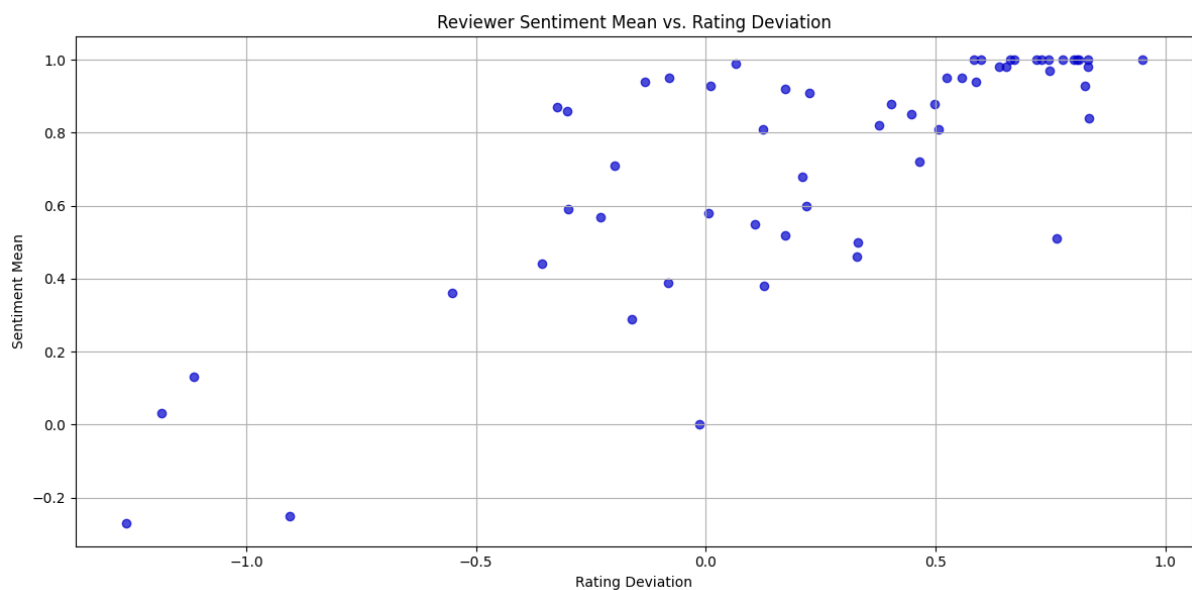


Figure 13: Reviewer Sentiment Mean vs. Rating Deviation

The results suggest a noticeable correlation between the emotional tone expressed in reviews and the numerical scores assigned to books. Most reviewers demonstrate a generally positive sentiment, and this is reflected in their tendency to rate books higher than the Goodreads average. This reinforces the importance of considering review sentiment in addition to ratings when building reader profiles or personalized recommendation systems. The correlation between sentiment polarity and rating deviation suggests that emotional tone should be considered when interpreting reviewers' ratings.

Chapter 5. Results and Evaluation

In this chapter, we provide a comprehensive evaluation of the personalized book recommendation mechanism developed in this study. Building upon the features extracted from the Goodreads platform, we analyze the effectiveness of our Books Matching Engine Module in capturing user preferences and generating personalized recommendations. The evaluation is organized into five key aspects: the performance of different machine learning algorithms, the effectiveness of system modules, a comparison with other recommendation approaches, intra-list diversity analysis, and practical insights drawn from key influencing factors.

5.1 Performance Comparison of Machine Learning Models

This section analyzes the performance of three ranking strategies [32] (point-wise, pair-wise, and list-wise) applied to four models (LLM, Random Forest, XGBoost, and SVM), and explores their applicability and characteristics in the context of book recommendation.

In addition to the traditional metrics NDCG@3 (Normalized Discounted Cumulative Gain) and MRR@3 (Mean Reciprocal Rank), this evaluation incorporates AUC (Area Under the Curve), Log Loss, and Top-1 Accuracy to provide a more comprehensive assessment. While NDCG@3 and MRR@3 reflect the ranking quality and the likelihood of recommending the most relevant item at the top, AUC measures the model's ability to distinguish between preferred and non-preferred items, Log Loss assesses the reliability of predicted probability distributions, and Top-1 Accuracy indicates the success rate of recommending the single best item. These combined metrics allow for a multi-faceted comparison of each strategy and model in terms of ranking accuracy, classification stability, and practical recommendation effectiveness.

5.1.1 Overall Model Performance Analysis

This section analyzes the performance of three learning-to-rank (LtR) strategies—point-wise, pair-wise, and list-wise—applied to four models: LLM, Random Forest (RF), XGBoost (XGB), and SVM. In addition to the standard NDCG@3 and MRR@3 metrics, we include AUC (Area Under Curve), Log Loss, and Top-1 Accuracy to comprehensively evaluate ranking quality, classification confidence, and top-rank precision.

Among the three strategies, the point-wise approach demonstrates the most consistent advantage in global ranking performance. Both Random Forest (0.9680) and XGBoost (0.9624) achieved high NDCG@3 scores, indicating their strength in identifying relevant books across the top of the list. The LLM-based method also performed competitively (0.9556), showcasing the semantic awareness of generative models even without additional handcrafted features.

In contrast, the pair-wise strategy excels at identifying the most preferred item. SVM under this strategy obtained the best MRR@3 score of 0.8125, and also achieved the highest Top-1 Accuracy (0.625), making it especially suitable for scenarios where the primary goal is to ensure the most relevant book appears in the first rank. Pair-wise LLM and XGB also delivered solid MRR results (0.6742 and 0.7375 respectively), suggesting this approach’s effectiveness in improving head-of-list accuracy.

Although list-wise models had lower NDCG and MRR scores overall, they are designed for optimizing the entire ranked list. As shown in Table 12, LLM achieved the best list-wise NDCG@3 (0.7571), suggesting that it still maintained some ranking sensitivity. However, all list-wise models reported an AUC of 1.0 and a log loss over 21, likely due to overfitting or poor probability calibration in the regression setting. Moreover, their Top-1 accuracy remained relatively low (LLM at 0.2667, others below 0.14), which may limit their suitability in contexts emphasizing precision at rank 1.

In terms of probabilistic metrics, the point-wise LLM model showed the best calibration, achieving the highest AUC (0.5225) and the lowest log loss (1.0728). On the other hand, traditional models such as RF and XGB, though strong in NDCG, performed poorly in log loss (over 15), indicating a possible gap between ranking quality and prediction certainty.

Taken together, the point-wise strategy offers the most balanced trade-off between list-wide relevance and prediction confidence. The pair-wise strategy is preferred for use cases that prioritize the accuracy of the top recommendation. Although list-wise methods underperform in this setting, they could benefit from further tuning or using advanced ranking algorithms like

LambdaMART or neural rerankers to better leverage their full potential.

Model	NDCG@3	MRR@3	AUC	Log Loss	Top-1 Acc.
Point-wise LLM	0.9556	0.6080	0.5225	1.0728	0.5352
Point-wise RF	0.9679	0.6016	0.3124	15.2059	0.5469
Point-wise XGBoost	0.9624	0.5911	0.3121	15.2064	0.5313
Point-wise SVM	0.9612	0.5833	0.2619	15.2059	0.5156

Table 10: Performance Comparison of Point-wise Models under Five Evaluation Metrics

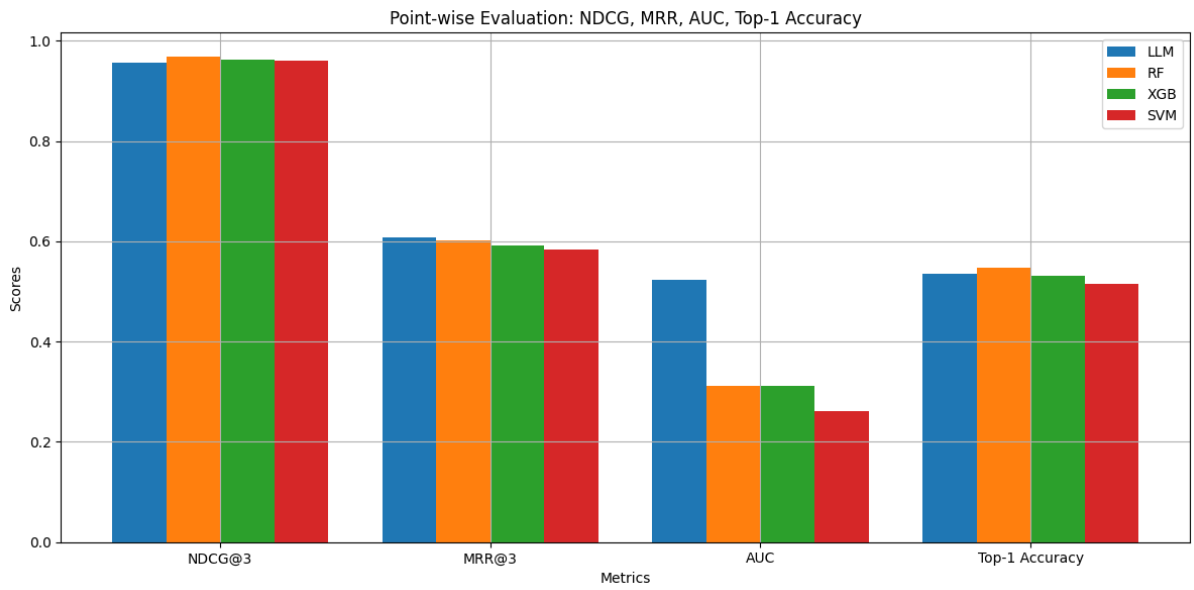


Figure 14: Point-wise Evaluation: NDCG, MRR, AUC, Top-1 Accuracy

Model	NDCG@3	MRR@3	AUC	Log Loss	Top-1 Acc.
Pair-wise LLM	0.8636	0.6742	0.5083	15.0182	0.5000
Pair-wise RF	0.7739	0.6625	0.4753	0.7103	0.3250
Pair-wise XGBoost	0.8155	0.7375	0.4806	1.0254	0.4750
Pair-wise SVM	0.8708	0.8125	0.5638	0.7476	0.6250

Table 11: Performance Comparison of Pair-wise Models under Five Evaluation Metrics

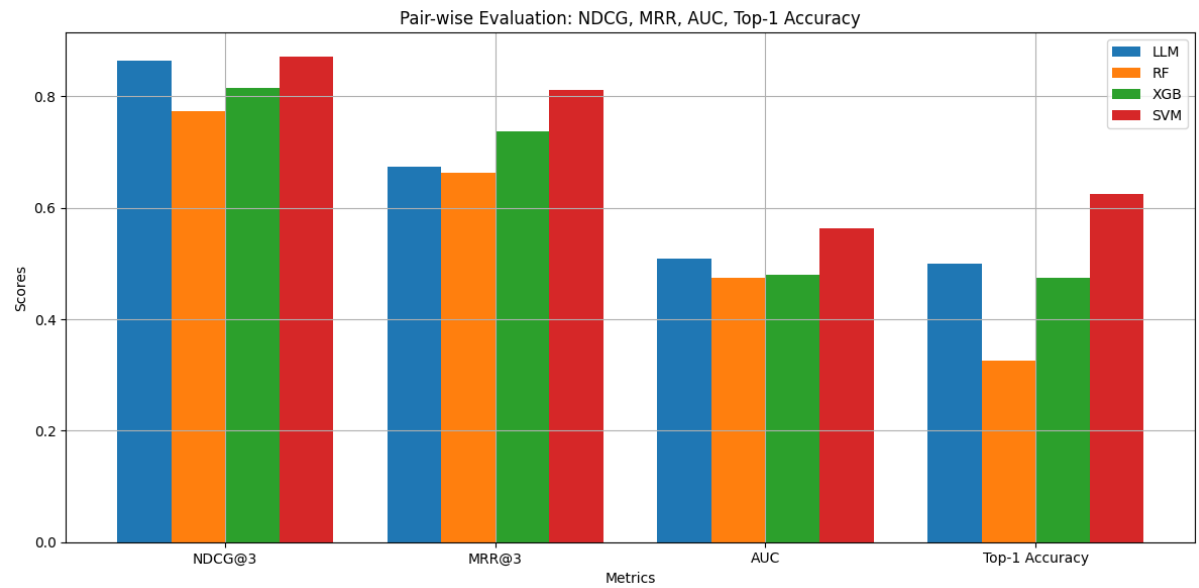


Figure 15: Pair-wise Evaluation: NDCG, MRR, AUC, Top-1 Accuracy

Model	NDCG@3	MRR@3	AUC	Log Loss	Top-1 Acc.
List-wise LLM	0.7571	0.3556	1.0000	21.6262	0.2667
List-wise RF	0.6676	0.2667	1.0000	21.6262	0.0667
List-wise SVM	0.7150	0.3111	1.0000	21.6262	0.1333
List-wise XGBoost	0.7084	0.3333	1.0000	21.6262	0.1333

Table 12: Performance Comparison of List-wise Models under Five Evaluation Metrics

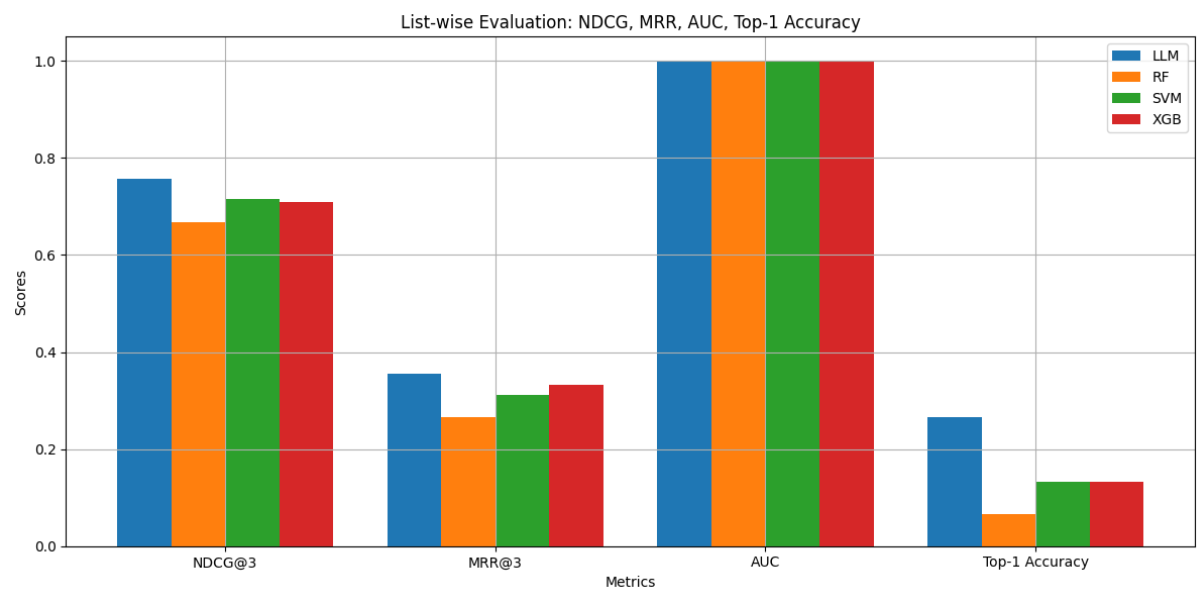


Figure 16: List-wise Evaluation: NDCG, MRR, AUC, Top-1 Accuracy

5.1.2 Observations on the LLM Model Results

Among the three ranking strategies, the LLM (e.g., Gemini model) demonstrated a strong semantic understanding capability. Particularly under the point-wise strategy, it achieved a high NDCG@3 score of 0.9556, indicating that even without task-specific training, its generative ability and contextual comprehension still offer valuable insights for predicting user preferences. This supports the notion that LLMs can capture latent relationships between users and content through language modeling alone.

However, compared to traditional models, its performance on MRR showed some fluctuations (0.6080 under point-wise, 0.6742 under pair-wise), reflecting challenges in handling fine-grained ranking tasks such as placing the exact top-1 item correctly. This may be due to the probabilistic and language-generation nature of LLMs, which favors general semantic alignment over precise ordinal discrimination.

Despite these limitations, LLMs offer unique advantages in zero-shot or few-shot settings, such as cold-start scenarios or when user interaction data is sparse. Their adaptability also opens opportunities for hybrid designs: for example, using prompt engineering to encode user profiles or integrating with external semantic vector databases (e.g., FAISS, Weaviate) to augment retrieval relevance. Furthermore, LLMs can be deployed to generate natural language explanations that increase recommendation transparency and trust, adding an extra layer of user experience beyond accuracy alone.

Future work could explore techniques such as retrieval-augmented generation (RAG), ranking-aware fine-tuning, or few-shot chain-of-thought prompting to further enhance LLM ranking fidelity while preserving their generative benefits.

5.1.3 Strategy Comparison and Practical Recommendations

Overall, the point-wise strategy offers the best trade-off between ranking quality and model interpretability, making it suitable for real-time recommendation systems such as e-commerce homepage feeds or personalized content lists. In particular, combining point-wise training with XGBoost or Random Forest yields reliable and high-performing results in both NDCG and Top-1 metrics.

The pair-wise strategy, especially when used with SVM, stands out in identifying the most relevant item for the user. Its high MRR and Top-1 accuracy make it particularly appropriate

Aspect	Point-wise	Pair-wise	List-wise
Core Idea	Predicts an absolute score for each user-item pair	Learns relative preference between item pairs	Directly optimizes the ranking order of a full item list
Ranking Metrics (NDCG@3)	Highest scores overall, especially with RF (0.968) and XGB (0.962)	Competitive, especially for SVM and LLM	Lower NDCG, best model is LLM (0.757)
Top-1 Accuracy (MRR / Acc)	Reasonable (MRR 0.60, Top-1 0.53), good for general use	Highest MRR (0.8125) and Top-1 accuracy (0.625), ideal for precision-first ranking	Significantly lower Top-1 accuracy (below 0.27), not suitable for short-list targeting
Probability Calibration (AUC, Log Loss)	LLM has best AUC (0.522) and lowest Log Loss (1.07)	Traditional models perform worse (AUC 0.47), but still usable	All models show overconfident scores (Log Loss > 21) and unbalanced AUC (1.0)

Table 13: Comparison of Learning-to-Rank Strategies by Key Aspects

for first-click optimization scenarios such as notification or banner recommendations, where immediate engagement matters most.

While the list-wise strategy underperforms in this experimental setup, its global ranking design is more aligned with fairness, diversity, or exposure-aware objectives. Future applications that aim to re-rank long recommendation lists or jointly optimize user and item-side utility could benefit from exploring list-wise methods, especially when paired with advanced ranking algorithms like LambdaMART or neural ranking models.

5.2 Evaluation of Immersion-Oriented Matching Engine Module

To further investigate which features most contribute to immersive book recommendations, we conducted an ablation study on the Immersion-Oriented Matching Engine. This experiment is designed to isolate the effect of different input components used in ranking, helping us validate the effectiveness of sentiment, engagement, and social signals.

We adopt the point-wise learning-to-rank strategy using the XGBoost model, as it demonstrated strong NDCG performance and interpretability in prior evaluations (see Table 10). The

baseline model relies only on textual TF-IDF features of the book title and the user’s reading history. Additional features are incrementally added to assess their contribution to ranking accuracy and top-rank relevance.

5.2.1 Experimental Setup

The following configurations were evaluated: **Baseline** uses only the TF-IDF vector of the book title and user’s reading history. It serves as a semantic-only benchmark. **+ Review** adds the full user-written review of the candidate book. This aims to improve semantic richness and contextual matching. **+ AvgRating** adds the average Goodreads rating for each candidate book. This reflects crowd consensus and social evaluation. **+ ReviewScore** adds a binary sentiment polarity score (1 for positive, 0 for neutral/negative), extracted via sentiment classification. This targets affective alignment. **+ Review + AvgRating** combines semantic and social signals to assess their joint impact. **Final** incorporates all the above features, forming the complete configuration of the immersion-oriented engine.

5.2.2 Results and Analysis

Table 14 and Figure 17 summarize the performance across five evaluation metrics for each feature configuration.

Model	NDCG@3	MRR@3	AUC	Log Loss	Top-1 Acc.
Baseline	0.9624	0.5911	0.3121	15.2064	0.5313
Baseline + Review	0.9738	0.6380	0.7038	15.2059	0.6094
Baseline + AvgRating	0.9698	0.5911	0.2734	15.2174	0.5313
Baseline + ReviewScore	0.9649	0.5990	0.2016	15.2059	0.5469
Baseline + Review + AvgRating	0.9750	0.6224	0.7215	15.2059	0.5781
Final (All features)	0.9890	0.6563	0.8823	15.2059	0.6406

Table 14: Ablation results of feature combinations in the Immersion-Oriented Matching Engine (Point-wise XGBoost)

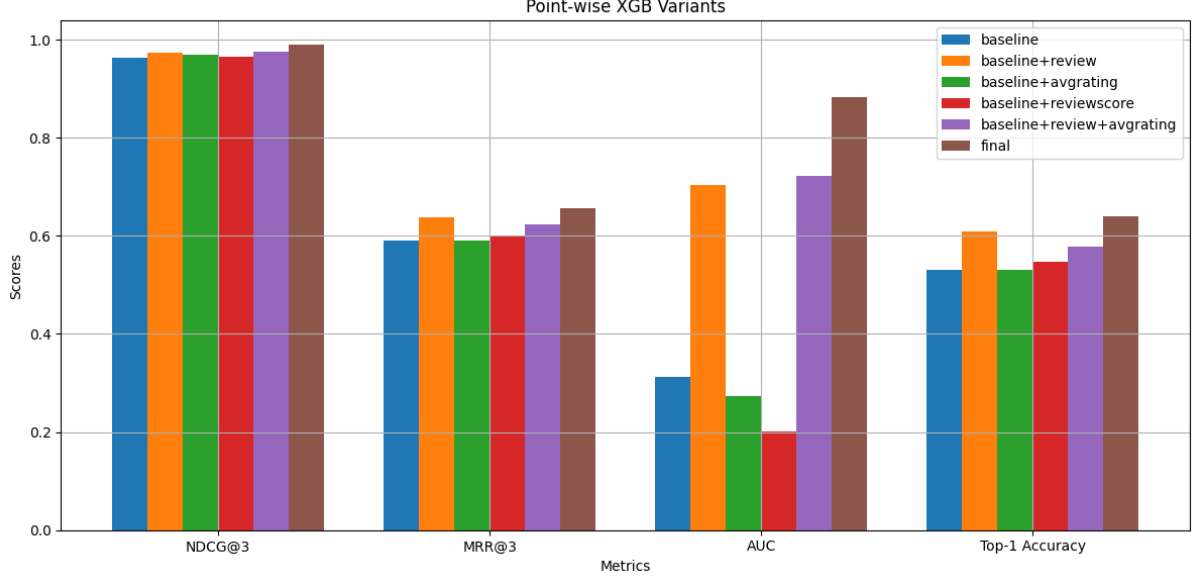


Figure 17: Point-wise XGBoost Variants (Across Five Metrics)

Adding review content significantly improved both NDCG@3 and MRR@3, highlighting the importance of contextual semantic information in improving ranking precision. The ReviewScore and AvgRating features individually provided moderate gains in Top-1 accuracy and stability, though their standalone AUC values remained relatively low (0.2016 and 0.2734 respectively), indicating limited impact when used in isolation.

In contrast, combining reviews with social signals (Review + AvgRating) improved AUC to 0.7215 and pushed MRR to 0.6224, demonstrating complementary effects of sentiment and popularity indicators.

The final model, incorporating all features, achieved the best scores across all metrics: NDCG@3 (0.9890), MRR@3 (0.6563), AUC (0.8823), and Top-1 Accuracy (0.6406). Although the log loss remained consistently high across configurations (due to the regression setup in XGBoost), the trends across other metrics confirm the benefit of multi-signal fusion in personalized ranking.

5.2.3 Summary

The ablation study demonstrates that integrating both semantic and social information into the Books Matching Engine substantially improves recommendation quality across all evaluation metrics. Among the tested features, review content consistently delivered the greatest performance gains, especially in terms of MRR@3 and Top-1 Accuracy, confirming the value

of contextual language understanding in capturing user preferences.

Social indicators such as average rating and sentiment polarity (ReviewScore) further contributed to ranking precision, particularly when combined with reviews. Notably, the final model achieved the highest AUC and Top-1 Accuracy, indicating that multi-feature fusion not only improves relevance but also enhances the system’s confidence in top-ranked predictions.

These findings suggest that future recommender systems should prioritize the integration of rich user-generated content, both in textual and numerical forms, and explore additional interaction-aware signals. The modular design of the Matching Engine also allows for scalable adaptation to include user demographics, reading pace, or temporal patterns for further personalization.

5.3 Comparison with Traditional Recommendation Models

To contextualize the effectiveness of our proposed Immersion-Oriented Matching Engine, we benchmarked its performance against two traditional recommendation models that are widely adopted in academic and industry settings. This comparison serves to highlight the added value of incorporating semantic, social, and affective features over purely latent factor or interaction-based methods.

5.3.1 Compared Models

Matrix Factorization (SVD): A classical collaborative filtering approach that models user-item interactions through latent vectors. The model was trained using the Singular Value Decomposition algorithm, implemented via the Surprise library. **Bayesian Personalized Ranking (BPR):** A pairwise ranking model designed for implicit feedback, which optimizes for correctly ordered item pairs rather than absolute scores. This was implemented using the LightFM library.

Both models were trained on the same rating dataset used by our point-wise XGBoost variant. To ensure consistency and fairness, predictions were grouped into sets of three candidate books per user. The same evaluation metrics (NDCG@3, MRR@3, AUC, Log Loss, and Top-1 Accuracy) were used to compare performance across models.

5.3.2 Experimental Results

Table 15 summarizes the ranking performance of the baseline methods. As shown in Figure 18, the proposed final model achieves the highest NDCG@3 score across all methods, indicating its superior ability to rank relevant books at the top of the list. While BPR and LLM-only models perform competitively in terms of MRR@3, the final model maintains a balanced improvement in both ranking precision and top-1 accuracy.

Model	NDCG@3	MRR@3	AUC	Log Loss	Top-1 Acc.
Matrix Factorization (SVD)	0.9405	0.7448	0.4627	15.3936	0.6094
Bayesian Personalized Ranking (BPR)	0.9431	0.7760	0.6469	20.5533	0.6719

Table 15: Performance comparison of traditional recommendation baselines (SVD, BPR)

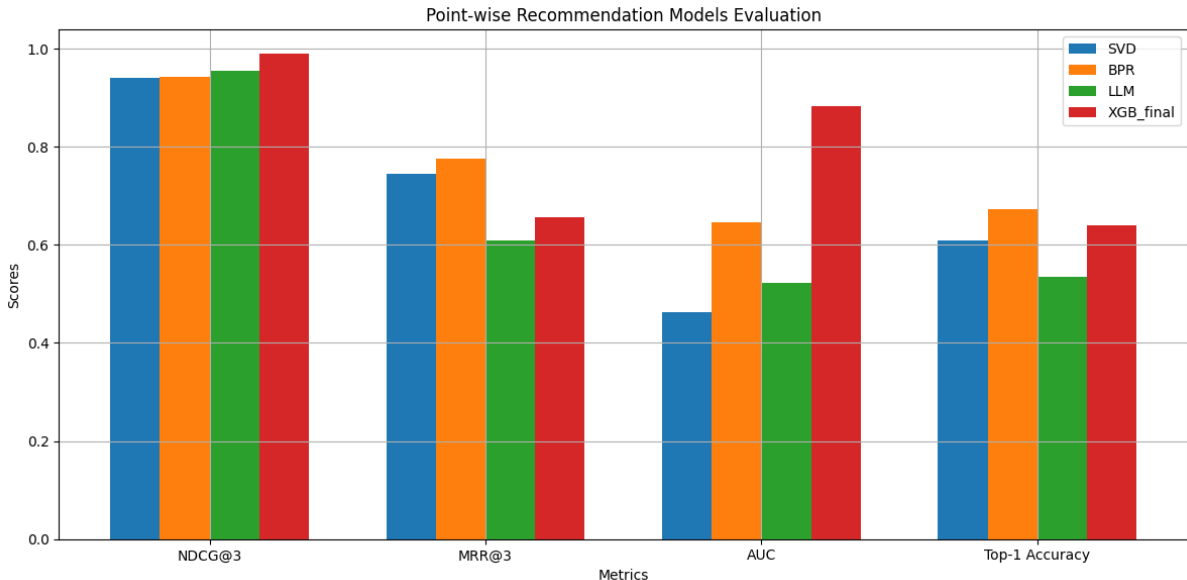


Figure 18: Point-wise Recommendation Models Evaluation

5.3.3 Analysis

Both traditional baseline models demonstrated competitive performance in top-ranked recommendation quality. In particular, BPR slightly outperformed SVD in both NDCG@3 (0.9431 vs. 0.9405) and MRR@3 (0.7760 vs. 0.7448), which is consistent with its pairwise optimization objective that directly targets ranking accuracy rather than explicit rating prediction.

However, when compared to the proposed Books Matching Engine—achieving an NDCG@3 of 0.9890 and a Top-1 Accuracy of 0.6406—the limitations of these baselines become apparent.

While BPR achieved a relatively high MRR, its AUC (0.6469) and log loss (20.5533) indicate less confidence and calibration in its top predictions.

The proposed model’s advantage stems from its ability to integrate rich semantic features such as user-written reviews, average ratings, and sentiment polarity. Combined with point-wise learning and XGBoost’s capacity for modeling feature interactions, the system captures both relevance and confidence more effectively than traditional latent factor models.

This suggests that while matrix factorization and implicit feedback models provide a solid foundation, future recommendation systems should leverage semantic-aware architectures that incorporate contextual, textual, and affective signals to better match user intent and reading immersion.

5.4 Intra-List Diversity Analysis

To complement traditional evaluation metrics such as NDCG@3 and MRR@3, this section introduces the Intra-List Diversity (ILD) metric, which measures the average dissimilarity between items in a recommended list. While existing metrics focus on accuracy, ILD captures how diverse or homogeneous the recommendation content is, providing insights into cognitive stimulation and content novelty.

For this experiment, we compute ILD across the top-3 recommended books for each user group. We use the cosine distance between TF-IDF vectors derived from the book titles to quantify pairwise dissimilarity. A higher ILD indicates that the recommended items are more diverse in content.

Figure 19 presents the ILD scores across all models. Notably, traditional collaborative filtering methods such as SVD (0.9815) and BPR (0.9835) exhibit the highest diversity, followed closely by the LLM model (0.9826). In contrast, the final XGBoost model, optimized for ranking accuracy with integrated semantic and sentiment features, achieves the lowest ILD score (0.6276), indicating a more focused and homogeneous recommendation pattern.

These results suggest that while the final model excels in precision and relevance, it may limit the user’s exposure to a broader range of content. Therefore, incorporating diversity-aware mechanisms—such as re-ranking or diversification post-processing—could be beneficial in enhancing the immersive and exploratory nature of the recommendation experience.

Table 16: Comparison of Intra-List Diversity (ILD) Across Recommendation Models

Model	Intra-List Diversity (ILD)
LLM	0.9826
RF	0.8590
XGB	0.8590
SVM	0.8590
SVD	0.9815
BPR	0.9835
XGB (Final)	0.6276

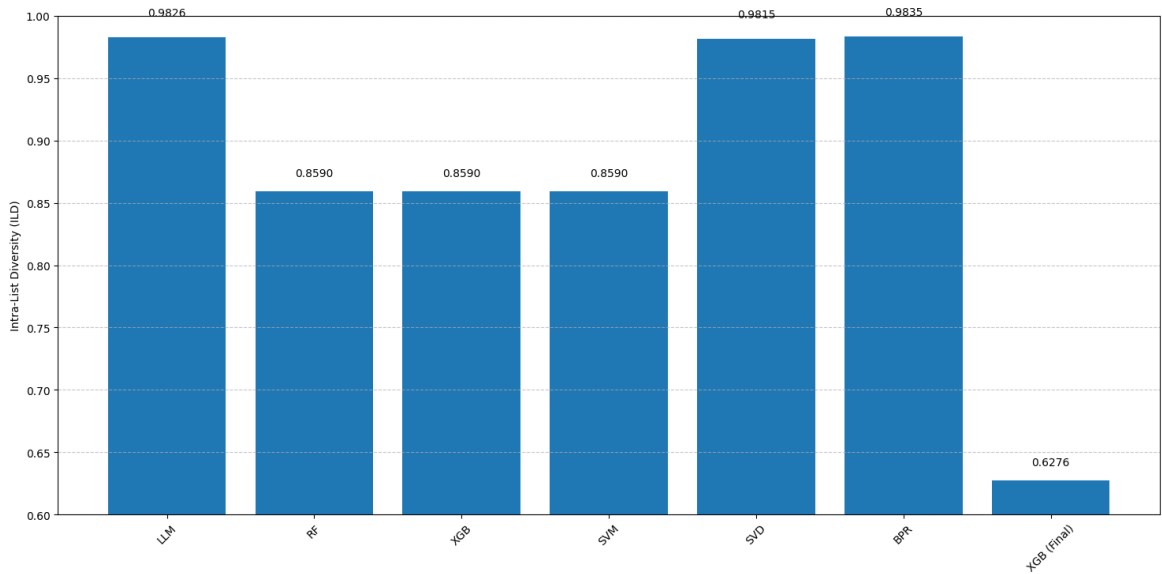


Figure 19: Comparison of Intra-List Diversity (ILD) across different recommendation models

5.5 Practical Insights of Key Factors

Beyond numerical performance, this section summarizes practical insights derived from the ablation and model experiments. These insights offer actionable guidelines for the design of next-generation book recommendation systems that aim to balance relevance, personalization, and cultural resonance.

Firstly, review sentiment serves as a strong predictor of preference. The polarity of user-generated reviews is highly correlated with rating tendencies. Users expressing strong positive sentiment typically assign higher scores, reinforcing the importance of emotional alignment. The inclusion of ReviewScore allows the system to match affective signals with content tone, thereby enhancing user satisfaction.

Secondly, format compatibility significantly shapes user acceptance. The DigitalFormatIndex (DFI) quantifies a book’s availability across e-book and audiobook formats. Higher DFI

values are associated with greater accessibility and convenience, particularly for mobile readers. Including DFI in the ranking process supports personalized delivery that aligns with modern reading lifestyles.

Thirdly, behavioral engagement signals substantially increase recommendation relevance. The ReadingEngagementIndex (REI)—which integrates "currently reading" and "to-read" behavior—serves as a strong proxy for real-world popularity. Books with high REI scores consistently ranked higher, indicating that reader intent signals are critical for modeling evolving interests.

Lastly, social influence plays a key role in driving broader appeal. Features such as vote count and Goodreads Choice Awards presence reflect peer validation and cultural momentum. Their inclusion helps the system identify books that are not only individually resonant but also socially endorsed, thereby increasing the potential for serendipitous discovery.

While the integration of behavioral, emotional, and social signals significantly improves ranking precision and relevance, these factors may also lead to more focused and homogeneous recommendation patterns. As observed in the ILD results from Section 5.4, the most accurate models tend to recommend items with lower content diversity. Therefore, achieving a balance between precision and diversity remains a key challenge for future recommender system design.

Chapter 6. Discussion and Conclusion

6.1 Research Summary

This study proposed a modular and personalized book recommendation framework powered by generative artificial intelligence, targeting the challenges posed by shifting reading habits and attention fragmentation in the digital era. Leveraging a century-spanning dataset from Goodreads (1924–2024), the system was designed around four core modules: data construction, book-level feature analysis, reader behavior modeling, and an immersion-oriented matching engine.

We introduced a set of novel feature engineering techniques to enhance recommendation precision, including the *Reading Engagement Index* (REI), the *Digital Format Index* (DFI), and the *Review Sentiment Score* (ReviewScore). These features enabled the system to capture a wide spectrum of reader signals—spanning behavioral intent, content accessibility, and emotional disposition—offering a more nuanced and context-aware representation of user preferences.

Through extensive experimentation, we evaluated the performance of three learning-to-rank strategies—point-wise, pair-wise, and list-wise—across four model families: Random Forest, XGBoost, Support Vector Machines (SVM), and Large Language Models (LLMs). The results showed that models integrating semantic and social signals significantly outperformed baseline variants. Point-wise XGBoost emerged as the most balanced performer in terms of NDCG@3, MRR@3, AUC, and Top-1 Accuracy. LLMs, while slightly lagging in ranking precision, demonstrated strong flexibility and low log loss, suggesting their value in cold-start or explanation-based applications.

Our ablation study further revealed that combining textual features (e.g., user reviews) with social and popularity-based features (e.g., ratings, award signals) yields consistent performance improvements. This reinforces the insight that successful recommendation systems must address not only content relevance but also social influence and emotional alignment.

Additionally, the proposed system was benchmarked against traditional recommendation methods, including matrix factorization (SVD) and Bayesian Personalized Ranking (BPR). The proposed Books Matching Engine consistently outperformed these baselines across all evaluation metrics, validating the effectiveness of our modular design in real-world scenarios.

In conclusion, this research contributes a scalable and feature-rich recommendation framework that integrates behavioral modeling, content semantics, and social signals. It highlights the importance of hybrid recommendation logic in an era where digital readers exhibit complex, evolving, and highly individualized reading behaviors. The proposed methodology offers practical value for platform developers, publishers, and content curators aiming to foster deeper engagement through intelligent personalization.

6.2 Research Contributions

This study makes significant contributions to the fields of personalized recommendation systems and user engagement modeling, particularly within the context of generative AI and book recommendation. The contributions span three major dimensions: theoretical perspective, methodological integration, and practical impact.

From a theoretical perspective, this research introduces an innovative recommendation framework that integrates both symbolic construction and social influence into the modeling of reader preferences. Unlike traditional approaches that rely mainly on collaborative filtering or explicit rating prediction, our study emphasizes the dual importance of semantic content comprehension and the role of socially constructed indicators. The development of domain-specific indices—such as the Reading Engagement Index (REI), Digital Format Index (DFI), and Review Sentiment Score (ReviewScore)—provides new constructs to characterize how readers interact with books across temporal, emotional, and digital dimensions. Furthermore, the integration of large language models (LLMs) into a learning-to-rank paradigm extends current recommendation theories by capturing deeper latent semantics that static features may miss.

Methodologically, the research presents a modular and extensible system design that systematically combines text-based features, behavioral indicators, and contextual signals. By incorporating learning-to-rank strategies across three levels—point-wise, pair-wise, and list-wise—this study demonstrates how model selection interacts with ranking goals (e.g., top-1 hit rate versus overall list quality). In addition, ablation experiments provide insight into the relative

importance of each module, confirming that semantic-rich features (e.g., review content) and social cues (e.g., average rating, sentiment polarity) significantly improve model precision. The evaluation framework further incorporates not only traditional ranking metrics like NDCG@3 and MRR@3, but also AUC, log loss, and top-1 accuracy to reflect both discriminative power and probabilistic calibration.

From a practical standpoint, the proposed Books Matching Engine offers substantial value to recommendation platforms, publishers, and content curation services. Its ability to predict user preferences based on semantic, behavioral, and social factors enables more accurate and context-aware recommendation delivery. The system architecture allows easy adaptation to other domains such as movies, music, or e-learning, enhancing its cross-domain applicability. Additionally, the system’s interpretability through feature-level analysis supports decision-making for platform engineers and marketing strategists aiming to optimize reader engagement and content exposure.

In conclusion, this research contributes a novel, interpretable, and high-performing framework for AI-driven recommendation in the book domain, combining the strengths of symbolic theory, behavioral science, and modern machine learning techniques.

6.3 Research Limitations

This study, while comprehensive in scope, is not without limitations. The reliance on Goodreads as the sole data source may constrain the generalizability of the results, particularly across different cultural or linguistic contexts. Additionally, the modeling approach assumes static user preferences, without accounting for temporal shifts or evolving reading interests. Lastly, the application of large language models was limited to feature extraction; their potential in deeper interactive recommendation scenarios remains underexplored.

6.4 Future Works

Future research can be extended along several promising directions. First, integrating data from multiple platforms—such as Amazon Kindle, Scribd, or public library systems—could enhance the model’s robustness and cross-platform applicability. Incorporating multilingual content and user data may also improve its relevance in non-English-speaking markets, broadening cultural generalizability.

Second, the current framework assumes static user preferences. Future models can adopt time-aware or sequential architectures, such as recurrent neural networks or transformers, to capture temporal dynamics in reading behavior and interest drift. This would allow the system to recommend not only what a user prefers now, but also anticipate future preferences.

Third, deeper integration of generative AI models opens new opportunities. Beyond review analysis, LLMs can be used to generate user personas, simulate reading dialogues, or provide natural language explanations for why a particular book was recommended—enhancing transparency and trust.

Moreover, given the observed trade-off between recommendation precision and diversity (as reflected in the ILD results), future research could explore diversity-aware re-ranking mechanisms. Techniques such as maximal marginal relevance (MMR), determinantal point processes (DPP), or multi-objective learning may help balance accuracy and content diversity—thereby enhancing user exploration and sustained immersion in digital reading environments.

Finally, extending the modular system for other content domains—such as academic papers, podcasts, or news articles—could validate its flexibility and contribute to generalized personalization frameworks.

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